

Catapult Project

Abstract

The purpose of this project was to design build and test a catapult. This catapult had to launch half of a cork a distance of 12 feet and a distance of 8 feet while remaining in an 8" x 8" x 8" inch cube. It was built using a mix of wood and 3D printed parts. Using mathematical models, the trajectory of the catapult was predicted and graphed to ensure the catapult's ability to hit its distance goals. The 12-foot mark was hit at a 52-degree launch angle, and the 8-foot mark was hit at a 76-degree angle. The catapults accuracy decreased as the range grew and increased with closer range. While not hitting the 12-foot mark perfectly like it did the 8-foot mark it still hit within an acceptable range of 12 feet.

Table of Contents

Abstract	2
Table of Figures	4
Table of Tables	4
Table of Equations	4
Introduction	5
Background Research	5
Preliminary Design	8
Solution 1	9
Solution 2	10
Solution 3	11
Justification of Best Idea	12
Results	12
Discussion	13
Conclusion & Recommendations	13
Appendices	14
Appendix A: Requirements	14
Appendix B: Conceptual Design	15
Appendix C: Embodiment Design	18
Appendix D: Prototyping	20
Appendix E: Dynamics Calculations	22
Appendix F: MATLAB/Excel Calculations and Graphs	24
Appendix G: Solidworks Model	29
Appendix H: Drawing Package	29

Table of Figures

Figure 1: Researched Catapult Design One	5
Figure 2: Researched Catapult Design Two	6
Figure 3: Projectile Trajectory Chart	6
Figure 4: Researched Catapult Design Three	8
Figure 5: Sketch of First Catapult Solution	9
Figure 6: Image of Second Catapult Solution.....	10
Figure 7: Image of Third Catapult Solution.....	11
Figure 8: Catapult Angles	20
Figure 9: Catapult Arm	21
Figure 10: Page One of Hand Calculations.....	22
Figure 11: Page Two of Hand Calculations.....	23
Figure 12: MATLAB graph showing the range various launch angles and numbers of rubberbands have	26
Figure 13: Trajectory of the Best Calculated Launch	27
Figure 14: Snip of an Excel sheet giving the calculated trajectory of the catapult.....	27
Figure 15: Excel graph of the catapults calculated trajectory	28
Figure 16: Final Catapult CAD Model	29

Table of Tables

Table 1: Weighted nalysis of the three solutions	15
Table 2: Pairwise comparison of the expected launch distance of the three solutions	16
Table 3: Pairwise comparison of the ability to be tuned of the three solutions	16
Table 4: Pairwise comparison of the ease of fabrication of the three solutions	16
Table 5: Pairwise comparison of the structural stability of the three solutions	16
Table 6: Pairwise comparison of the degree of creativity of the three solutions	17
Table 7: Pairwise comparison's totals of the three solutions	17

Table of Equations

Equation 1: Projectile Motion: Range Equation	7
Equation 2: Torque Equation	7
Equation 3: Arc Length Equation	7

Introduction

Our catapult was inspired by existing simplistic catapult designs. We need to keep our fabrication process simple due to our limited methods of fabrication. We were very restricted on what tools and materials we were able to use. We were restricted to parts that are not machined, making tolerances hard to make precise. See appendix H for exact tolerances. This made any complex joints or surfaces out of the scope of possibilities. We ultimately chose our design to keep the fabrication process simple while maximizing our range and giving us the flexibility to easily modify the shot distance. See appendix C for further information on the fabrication process. This led to us choosing a compact frame with the ability to change the launch angle of the catapult. This gave us the most freedom to hit our target of 12 feet and 8 feet without the need for extensive tweaking of the launch speed.

Background Research

Our team has decided that we would like to do a catapult whose energy stems from an elastic band. While doing research, we saw a couple of ideas on how to build the catapult, with the image below as the main body idea.

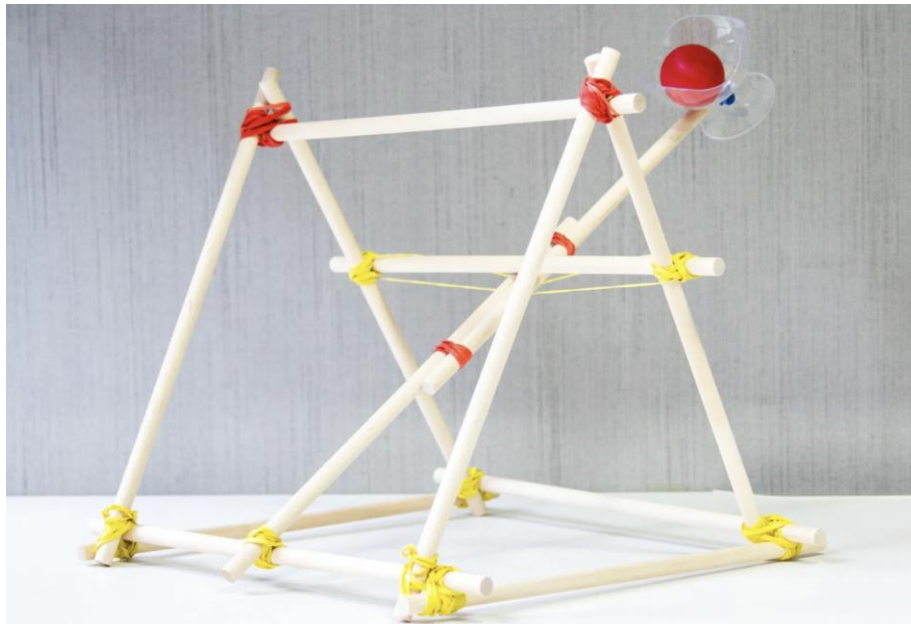


Figure 1: Researched Catapult Design One

In this image, we have a basic catapult that is easy to build, uses an elastic band effectively, and uses a pole to stop the arm of the catapult at a certain launch angle (just like what we had intended on). However, the above catapult seems very unstable from the sides and seems as if it will wobble while accelerating, which will create an odd and uncertain point of launch. Another con of this design is the way the rubber band is placed

on the catapults arm. With this placement you would lose over half of the potential energy that's possible from the rubber band.

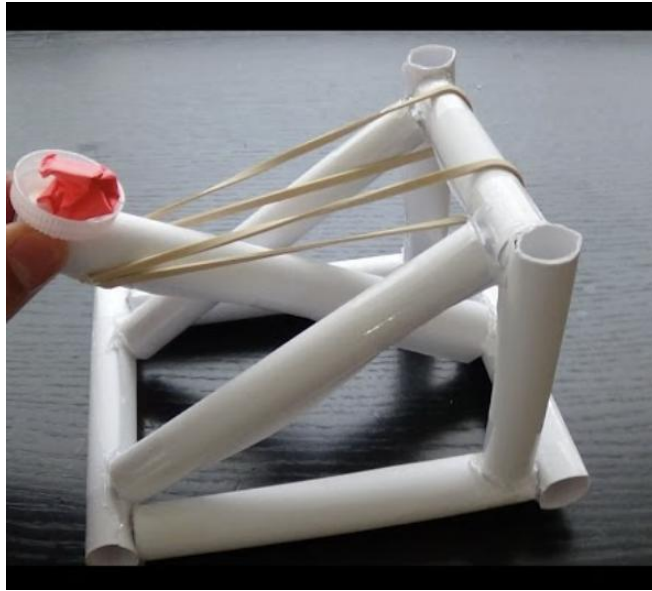


Figure 2: Researched Catapult Design Two

In this picture we can see that the catapults arm is built into a cylinder at the bottom of the catapult that rotates as the catapult is in motion, and this design is was a good solution to our catapult arms' instability issue. We will design the catapults arm to be 3d printed in a T-shape, with the cylinder as the top of the T-shape, thus will incorporate at the bottom of our catapult and we can guarantee that the catapult arm will not move side to side, all while the cylinder shape allows for complete 360 degree motion of the arm. The rubber bands orientation on the catapults is a large improvement on the power it can generate compared to the last catapults.

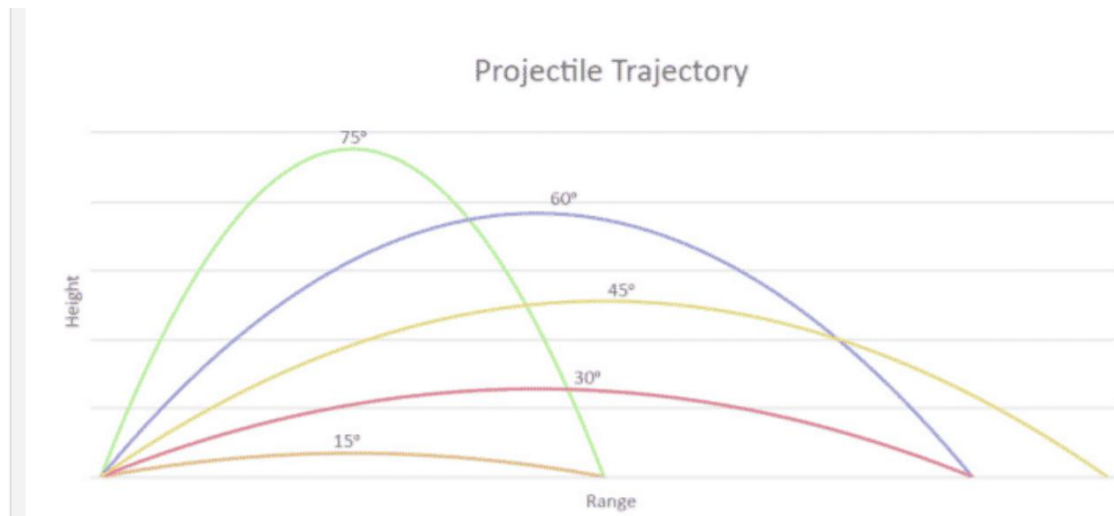


Figure 3: Projectile Trajectory Chart

Next, we needed to decide a proper launch angle, which while doing research we came across this graph from, <https://discovere.org/engineering-activities/design-a-catapult/>, with this information we decided that it would be best to use a launch angle between 30 and 45 degrees to hit our distance goal. This can be seen in the range equation.

Equation 1: Projectile Motion: Range Equation

$$\frac{v_0^2 \sin(2\theta)}{g},$$

As the sin of 90 is 1 and anything else between 0 and 180 is a number less than one the ideal launch angle is $90/2$ which comes out to 45 degrees. However, because the velocity on a catapult increases with the distance the catapults arm travels it is likely that the optimal launch for the farthest distance will be slightly after the 45-degree mark with its higher velocity.

To get full potential energy out of our rubber band, we need to decide where to place the rubber bands and the geometry of it that allows them to increase the speed of the arm that allows for the furthest throw. One potential placement of the rubber bands is higher up on the catapult so that way when stretched by the catapults arm, the rubber band has a high tension and will allow for a further shot range. This orientation will allow for the most force to be exerted onto the arm. Another possibility is for the rubber bands to be mounted closer to the base of the catapult. This configuration allows the arm to accelerate much faster and end at a higher velocity. The rubber band has a much lower travel distance compared to the other methods of attaching it, letting the arm snap back faster given its rotation inertia is not too high. This is at the cost of torque. The lower the rubber band the less torque gets exerted, and if too little torque is exerted this will have major negative impacts on the catapults' throwing ability. This can be described with the torque equation.

Equation 2: Torque Equation

$$\tau = r \cdot F \cdot \sin(\theta)$$

This shows how the torque increases further up the arm the rubber band is placed and shows how the angle of the rubber band in relation to the arm can impact how much torque gets generated by the rubber bands. This in addition to the arc length equation which shows the linear distance the rubber band must travel gives us the ideal distance the rubber band must be on the arm.

Equation 3: Arc Length Equation

$$s = r \cdot \theta$$

Reducing the overall travel distance the rubber bands must contract by will increase the speed at which the end of the catapult arm is traveling given it is supplied enough torque to quickly accelerate. This change in acceleration was tested and prototyped with multiple rubber band placements in our catapult arm. Information on this can be found in appendix D.



Figure 4: Researched Catapult Design Three

While this configuration adds another layer of complexity, it allows for a further shot. With the additional force that is required to use this configuration, it needs to be considered in the 3D printed arm as there is a risk of snapping it in half from impact force or even from the higher forces exerted on its weak points. The more traditional way of attaching the bands at the top makes it much easier to control the force that the catapult is firing at. There is a middle ground between the two that is ideal for maximizing the distance thrown.

Preliminary Design

During the preliminary design phase, we developed three different concepts for our potential catapult design. Each design was based on our research and other existing catapult designs adapted to fit this project. All three designs follow below.

Solution 1

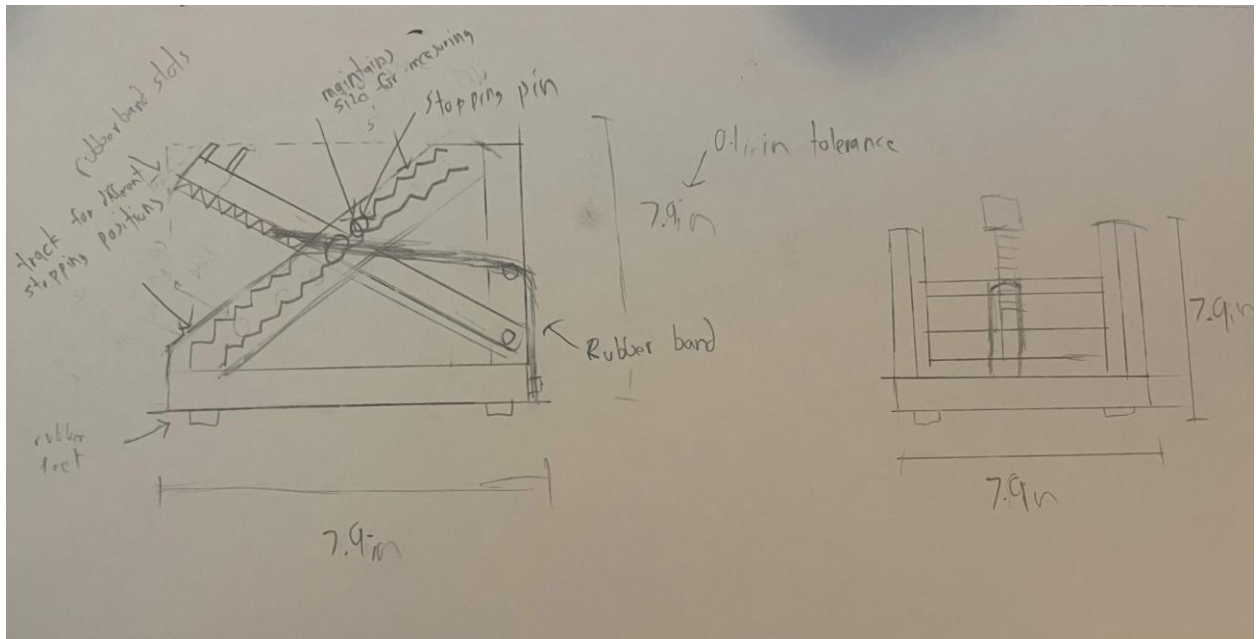


Figure 5: Sketch of First Catapult Solution

The rubber bands can change the spot they attach to the arm, allowing for more adjustment in firing power. The rubber bands are routed around a pin to allow for the bands to constantly keep tension while also ensuring it has a good angle to pull from to allow for a better force transfer. The staircasing track has a pin inserted into it that can sit on each level of the track. This allows for each step to be a different angle the catapult can fire at. The pin also is used to maintain its size requirement when being measured. It's built slightly under the 8-inch maximum to ensure any imperfections aren't out of the maximum size. The catapult is attached using a pin joint. The rubber feet help to keep the catapult stable when fired.

Solution 2

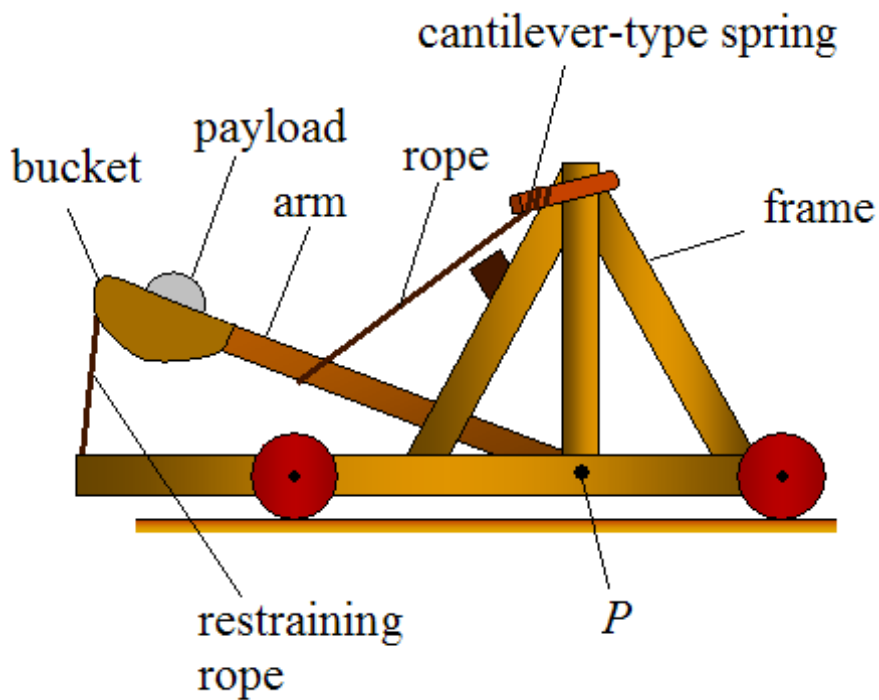


Figure 6: Image of Second Catapult Solution

This catapult design would use an A-frame support, providing a rigid triangular shape. Our design would not have wheels and would instead use a wide base to minimize movement of the platform when launching. The scoop or 'bucket' in the picture can be perforated to have small holes like a racket to reduce mass and rid of some resistance. A fixed block will stop the arm at the same angle every time to provide consistency. Rubber bands will be attached between the arm and the base. A long arm can be used to give the projectile additional velocity. This configuration gives this design power, accuracy, stability and simplicity in construction.

Solution 3



Figure 7: Image of Third Catapult Solution

This catapult design is ideal for maximum distance because it combines a long throwing arm, a high and well positioned pivot, strong triangular bracing, and a lightweight arm. The long arm increases tip velocity and torque, while the elevated pivot allows a larger swing arc and better launch angle. The triangular supports prevent frame flexing, so energy isn't lost to structural bending, ensuring more force transfers into the projectile. A lightweight arm minimizes wasted energy accelerating the arm itself, improving efficiency, and the cup holds the projectile securely for a consistent, controlled release. Together, these features maximize energy transfer, launch speed, and stability, resulting in greater overall range.

Justification of Best Idea

Our chosen catapult design, solution 1 found in appendix B, was our most optimal approach to the design problem. While it remains similar and inspired by existing solutions, it uses space more effectively than them. Many existing solutions have a triangle brace on each side of the main catapult tower. To ensure we can reach the 12 feet mark, the length of the throwing arm needs to be maximized. Any structure in front of the catapult limits the total length the arm can be. While we lose some overall structural strength, we gain a huge improvement in maximum distance from doing so. Many existing solutions also have rubber bands placed where this is low to no tension at the top of the arms swing. Our solution fixes the loss of power in the upper portion of the swing by keeping the rubber bands stretched even at its resting position.

Our chosen design will outperform our other two ideas. Our other two solutions both contain a lot of wasted space in front of the catapults arm. This is similar to existing solutions. Solution 3's structure is susceptible to twisting around the corners and becoming unstable during the build process. Both solutions also have no way to easily hit a secondary launch distance, nor have the ability to easily adjust the power aside from changing the rubber bands. See Appendix B for a full comparison of designs. This would make any difference between the calculated distance and actual distance far easier to adjust for.

This design satisfies all of the requirements. It uses the limited 8" x 8" x 8" maximum size to the full extent. It also ensures that both the secondary and primary distances are both able to be fired. A full list of requirements of the catapult can be found in appendix A. While each of the solutions hit almost every single requirement, our chosen design ensures that every single requirement is met, including the secondary distance. It also ensures that the secondary distance is hit without the addition or removal of any parts to the catapult.

Results

The final catapult design demonstrated the capability to achieve both the required distance and variable launch performance. During testing, the device consistently propelled the cork to the 8 ft target and successfully deposited it within the designated box, thereby satisfying the primary accuracy and distance requirements. When adjusted to maximize range, the catapult was capable of launching the cork approximately 12 ft. However, under these conditions, the projectile did not consistently land within the target box, indicating a reduction in accuracy at increased distances. The SolidWorks model, found in appendix G, and supporting calculations showed general agreement with experimental observations, particularly in illustrating the relationship between increased stored energy and extended launch distance.

Discussion

The experimental results were largely consistent with theoretical predictions. Analytical calculations indicated that increasing the stored potential energy in the system would result in greater projectile distance, which was validated by the 12 ft launch capability. However, the observed decrease in accuracy at extended distances highlights the influence of practical factors not fully accounted for in theoretical models. See appendix E and F for more information. These factors likely include minor variations in release angle, frictional losses at the pivot point, and inconsistencies in projectile positioning prior to launch. At the 8 ft distance, the catapult exhibited a high degree of reliability and precision, consistently meeting both accuracy and distance requirements. This suggests that the design performs optimally within a defined operational range. Conversely, at 12 ft, although the catapult achieved the required distance, it failed to meet accuracy constraints, indicating that increased energy input may negatively impact system stability and control. Overall, the catapult satisfied the majority of the design requirements, particularly those associated with the primary target distance, while revealing limitations in performance at extended ranges.

Conclusion & Recommendations

The catapult demonstrated effective performance within its intended operating range, particularly at the 8 ft distance where it consistently met both accuracy and distance criteria. This indicates that the design is stable, reliable, and well-suited for moderate energy conditions. Additionally, the system's ability to achieve a maximum range of 12 ft confirms that it can generate sufficient energy for extended launches. Despite these strengths, the reduction in accuracy at greater distances suggests the need for further design refinement. Future improvements could focus on enhancing the consistency of the release mechanism to ensure a uniform launch angle, reinforcing the structural components to minimize unintended movement, and incorporating guiding features to better control projectile trajectory. This project highlights the importance of balancing power and precision in mechanical design. It also demonstrates that while theoretical models provide valuable insight, real-world performance is influenced by additional variables that must be considered. Overall, the project successfully applied fundamental engineering principles and provided meaningful experience in design evaluation.

Appendices

Appendix A: Requirements

- Catapult must fit within an 8" x 8" x 8" cube when not in operation.
- Catapult must be one cohesive assembly.
- The throwing arm of the catapult must be made by additive manufacturing.
- All design decisions must be justified.
- The material choices must be justified and outlined.
- That catapult must fire one half of a standard size cork.
- The catapult must fire 12 feet.
- The catapult should also be able to fire a secondary distance of 8 feet without the addition or removal of parts.
- The catapult must be constructed without the use of machining.
- The catapult should use rubber bands as their main source of energy.

Appendix B: Conceptual Design

For our idea generation method, we went with one of the logical methods. We generated ideas by assessing other solutions that already exist. This is a good method for this project as catapults aren't a new concept, and there are many designs that can be used and learned from rather than starting from scratch. This is why using the method of assessing preexisting solutions and taking aspects from different designs and applying them to your own is a good method for this project.

For our idea evaluation methods, we chose to use both a weighted analysis and a pairwise comparison. A weighted analysis is good when comparing the overall designs of the catapults. It allows you to prioritize different categories as needed. It's great when having to pick one of the designs. The pairwise comparisons are good at directly comparing individual aspects of different designs. It visualizes which design is better in which areas. This can help improve the final design. It shows which aspects of which designs work best and can help take the best aspects of each design and bring them together into one complete design.

Weighted Analysis

	Weights	Solution 1	Solution 2	Solution 3
Expected launch distance	8	8	8	6
Ability to be tuned(corrections for overshoot/undershoot)	6	4	2	3
Ease of fabrication	5	4	5	8
Structural Stability	6	5	7	4
Degree of creativity	3	6	4	2
Totals		156	155	136

Table 1: Weighted analysis of the three solutions

Pairwise Comparisons

Expected launch distance	Solution 1	Solution 2	Solution 3
Solution 1	-	0	-1
Solution 2	0	-	-1
Solution 3	1	1	-
Totals	1	1	-2

Table 2: Pairwise comparison of the expected launch distance of the three solutions

Ability to be tuned (corrections for overshoot/undershoot)	Solution 1	Solution 2	Solution 3
Solution 1	-	-1	-1
Solution 2	1	-	1
Solution 3	1	-1	-
Totals	2	-2	0

Table 3: Pairwise comparison of the ability to be tuned of the three solutions

Ease of fabrication	Solution 1	Solution 2	Solution 3
Solution 1	-	1	1
Solution 2	-1	-	1
Solution 3	-1	-1	-
Totals	-2	0	2

Table 4: Pairwise comparison of the ease of fabrication of the three solutions

Structural Stability	Solution 1	Solution 2	Solution 3
Solution 1	-	1	-1
Solution 2	-1	-	-1
Solution 3	1	1	-
Totals	0	2	-2

Table 5: Pairwise comparison of the structural stability of the three solutions

Degree of creativity	Solution 1	Solution 2	Solution 3
Solution 1	-	-1	-1
Solution 2	1	-	-1
Solution 3	1	1	-
Totals	2	0	-2

Table 6: Pairwise comparison of the degree of creativity of the three solutions

	Solution 1	Solution 2	Solution 3
Expected launch distance	1	1	-2
Ability to be tuned(corrections for overshoot/undershoot)	2	-2	0
Ease of fabrication	-2	0	2
Structural Stability	0	2	-2
Degree of creativity	2	0	-2
Totals	3	1	-4

Table 7: Pairwise comparison's totals of the three solutions

Appendix C: Embodiment Design

The base of the catapult will be made out of a quarter inch plywood sheet that is 7.5 inches wide and 7.5 inches tall. The solid wood piece provides a solid base to work with that break under the forces of the catapult. Wood is also a piece we have on hand and won't require any additional expenses. The width and height provide some clearance to ensure that the catapult's body and anything slightly protruding from it maintain within the size range. The vertical pillars can be made out of $\frac{3}{4}$ inch square wooden dowels. These should be attached 2 inches in from two of the parallel edges and made flush with the perpendicular edge. The thicker wood helps strengthen the pillars, preventing them from breaking under load. These pieces should be attached using a mix of wood glue and screws. The screws are needed to prevent the piece from lifting upwards, and the wood glue is needed to prevent shear forces from breaking the piece off. The screw holes should be predrilled to prevent the wood from splitting. This piece is also ideal as it is provided and won't present external costs. The heavier pieces used in the base will help prevent the reactionary forces from causing recoil in the catapult. There also needs to be a small triangle in corner of the pillars to keep the pillars straight. These should be screwed with predrilled holes.

About $\frac{1}{2}$ inch up from the start of the vertical pillars a $\frac{3}{8}$ -inch hole should be drilled into the center of both beams. This hole may need to be made slightly larger using sandpaper or other tools to allow a $\frac{3}{8}$ -inch circular dowel to be inserted. Then a 3D printed sheet can be created about a centimeter thick that has holes large enough to fit the base of the screw but not its head. There should be one on each side of the structure attached using screw holes that attach to the base of the structure and the vertical pillars using the same method as previously used. The 3D print is needed as the slots will be created at angles that are calculated to hit the target distance. These angles are more precise when created by the 3D printer. These 3D prints then get supported by a $\frac{1}{8}$ th inch thick wood sheet that lines the 3D print. The fastener kits can be used to attach them together. This is also done on each side. The holes can be drilled into the wood using the 3D print as a guide. The wood pieces are needed as the 3D print is at risk of breaking from the impact force.

Above the holes in the vertical pillar another $\frac{1}{4}$ inch hole can be made on each side. A $\frac{3}{8}$ th inch round dowel can then be inserted into the hole and glued in place. This lets the rubber bands apply the force in the correct spot. $\frac{3}{8}$ th inch is needed as this part could be under a lot of lateral stress from the rubber bands. The piece to secure the rubber bands to the base can then be added centered with it. This can be a simple screw and nut going through the base with the rubber band wrapped around it. Rubber feet can then be added to each of the corners of the base. We can use erasers to do so as they are readily available.

The catapults throwing arm will be $\frac{3}{4}$ inch thick at the base of it $\frac{1}{2}$ tall. The edges will be rounded to prevent weak points. The base being flat helps with the 3D printing process. The one end will have a capsule slightly larger than the cork, and the other will have a circular ring slightly larger than the $\frac{3}{8}$ th inch shaft. It should be printed so that the ring is touching the build plate to keep it as circular as possible. The shaft will be about 7 inches long. Spacers with the same diameter will also be printed to prevent the catapult from shifting along the shaft. One spacer will be inserted onto the $\frac{3}{8}$ -inch-thick circular dowel after it is pushed through one of the holes. Then the catapult arm is inserted along with the other spacer. The edges of the dowel should be glued into the vertical pillars to prevent slop and unnecessary friction. The dowel should be first sanded down to reduce friction.

Appendix D: Prototyping

Much of our prototyping was done inside SolidWorks. The big two major aspects of the catapult that got prototyped in solid works were the angle the pin will stop the catapult's arm at. And the total length of the rubber band at the various stages of the catapult. While both use a non-physical model the digital model must be highly detailed as the numbers we get back from the prototype must be accurate to ensure that our mathematical models are correct. Both of these require the entire catapult system to be modeled.

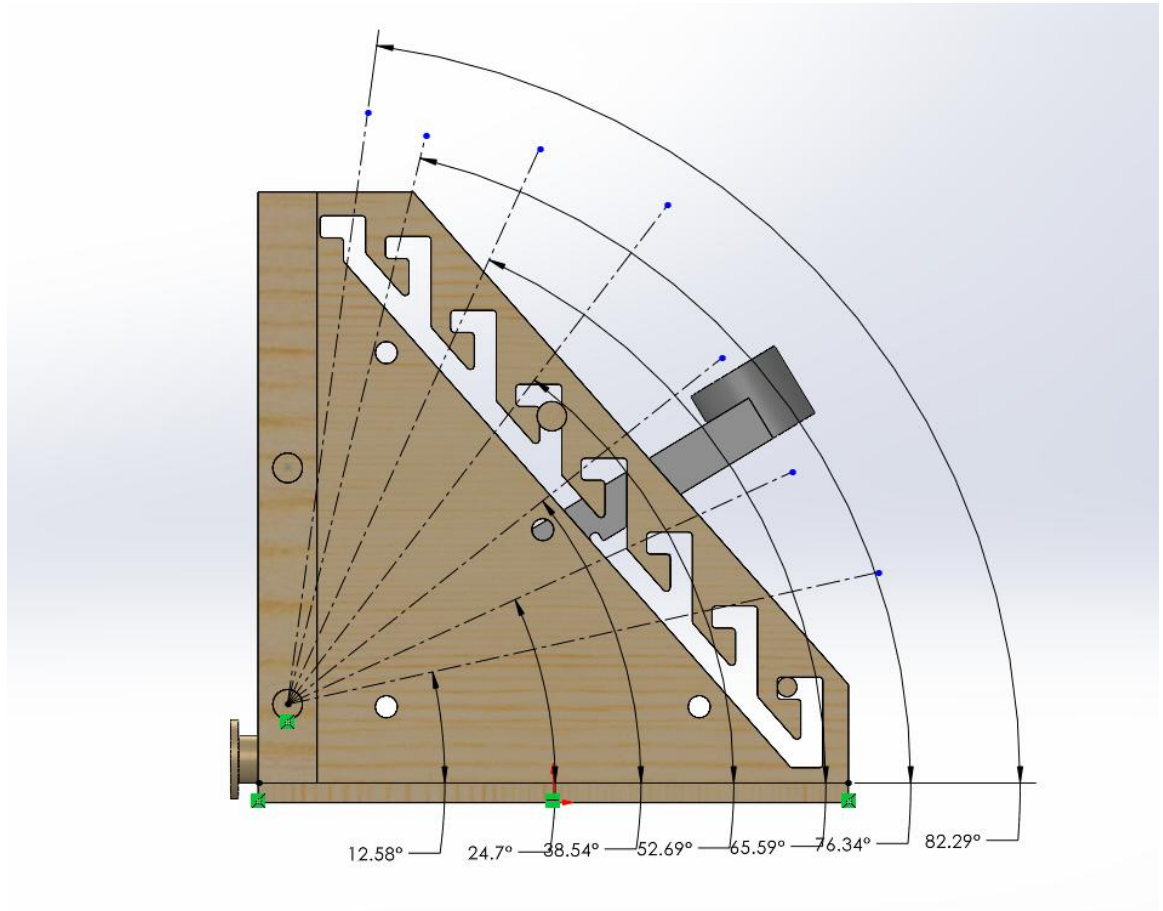


Figure 8: Catapult Angles

We also used a physical prototype in our catapults arm. Rather than relying on calculation to find the ideal rubber band placement our catapults arm was printed with multiple slots the rubber band can fit into. This allows us to test the ideal location of the rubber band's placement. This prototype was built into the final catapult arms design. Having to constantly reprint it at different lengths was too time-consuming for our allowed time and would increase costs.

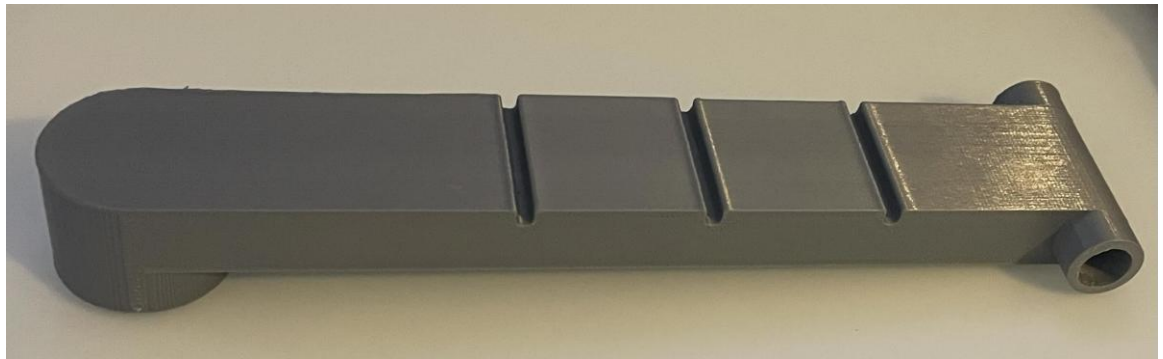


Figure 9: Catapult Arm

Appendix E: Dynamics Calculations

Ref

Catapult project Dynamics

- Energy stored (U_E)
$$U_E = \frac{1}{2} kx^2$$

x = how far we pull band back
 k = spring constant
- Move from potential to kinetic energy
$$\frac{1}{2} kx^2 = \frac{1}{2} m v_0^2$$

m = mass (kg) v_0 = initial velocity
- Find initial velocity
$$v_0 = \sqrt{\frac{kx^2}{m}}$$
- Find Maximum distance given the launch angle
$$X_m = \frac{v_0^2 \sin(2\theta)}{g}$$

g → gravity (9.81 m/s^2)
- Our Calculations
 - Average k for a band is 25
 - launch angle is maximized at 45°
 - Corollary is 2g • We will pull band back 5in
$$2g = .002 \text{ kg} \quad 5\text{in} = 0.127 \text{ m}$$
$$U_E = \frac{1}{2} (25) (.127)^2 \approx 0.2016 \quad X_m = \frac{14.149^2 \sin(2(45))}{9.81}$$
$$0.2016 = \frac{1}{2} (.002) v_0^2 \quad v_0 \approx 14.199 \quad X_m = 20.55 \text{ m}$$

— too far

Figure 10: Page One of Hand Calculations

$k=20$, $X_m=14\text{in}$ (included margin of error), $\theta=45^\circ$

$$0.3556\text{m} = \frac{v_0^2 \sin(2(45))}{9.81}$$

$$v_0 \approx 1.8677\text{m/s}$$

$$\frac{1}{2}(20)x^2 = \frac{1}{2}(0.002)(1.8677)^2$$

$$x \approx 0.18677\text{m} \approx 0.735\text{inches}$$

If we use a 45° launch angle,
assuming $k=20$, we can stretch the
arm back 14 in to reach our goal.

Figure 11: Page Two of Hand Calculations

Appendix F: MATLAB/Excel Calculations and Graphs

```
clc
clear all
close all
%Geometry variables
spring_constant = 100; %Estimate of spring constant of one rubber band
Rubber_Band_Length_inches = 3; %Length of rubber band in inches
Mass_cork_grams = 2; %Mass of the cork
Diameter_cork_inches = 1; %Diameter of the cork
Length_arm_inches = 7.03; %Length of arm from center to middle of cork holder

Moment_of_inertia_arm_inlb = 5.13; %Solidowork's calculated moment of inertia

Number_of_Bands = 2:2:10; %Number of rubber bands broken in even increments
Starting_Angle = 3.04; %Firing angle
Possible_Angles = [12.58, 24.7, 38.54, 52.69, 65.59, 76.34, 82.29]; %Possible launch angles
Rubber_Band_Starting_Length_inches = 8.59; %Firing rubber band length
Rubber_Band_Lengths_inches = [8.233, 7.74, 7.11, 6.52, 6.05, 5.52, 5.07]; %Rubber band length at launch angles

%Converting units
Mass_cork = Mass_cork_grams / 1000; % Convery mass of cork to kilograms
Lenth_arm = Length_arm_inches * 0.0254; % Convert Arm legnth to meters
Moment_of_inertia_arm = (Moment_of_inertia_arm_inlb * 0.0002926) + (Mass_cork * Lenth_arm^2); %Add cork after converting units
Rubber_Band_Length = Rubber_Band_Length_inches * 0.0254; % Convert length of band to meters
Rubber_Band_Starting_Length = Rubber_Band_Starting_Length_inches * 0.0254; % Convert initial stretch to meters
Rubber_Band_Lengths = Rubber_Band_Lengths_inches * 0.0254; % Convert Band lengths to meters
Cork_area = (pi * ((Diameter_cork_inches * 0.0254)/2)^2); %Factors in air drag

%Calculating and graphing
%Create figure
figure(1);
hold on;
colors = ['r', 'b', 'g', 'm', 'c'];

%For loop for number of bands
for k = 1:length(Number_of_Bands)
    bands = Number_of_Bands(k); %The number of bands for the loop #

    %For loop for each angle we can launch from
    for i = 1:length(Possible_Angles)
```

```

    %Calculating rubber band stretch
    Start_Stretch = Rubber_Band_Starting_Length - Rubber_Band_Length; %Tension
in band at start
    Stop_Stretch = Rubber_Band_Lengths(i) - Rubber_Band_Length;%Tension in band
at end

    %Potential energy
    PE = 0.5 * bands * spring_constant * (Start_Stretch^2 - Stop_Stretch^2); %Total
energy in band

    %Velocities
    Angular = sqrt((2* PE) / Moment_of_inertia_arm); %convert potential energy to
angular velocity
    linear = Angular * Lenth_arm; %Convert angular velocity to linear velocity

    %Projectile motion
    launch = Possible_Angles(i); %Possible release angles for the in the loop
    eqs = [0; linear*cosd(launch); 0; linear*sind(launch)];
    increment = 0:0.01:10;
    [t, sol] = ode45(@(t,y) projectile_drag(t, y, Mass_cork, Cork_area), increment, eqs);
    %Solving differential equation accounting for air drag

    %Ground impact
    impact = find(sol(:,3) < 0, 1, 'first'); %finds the point where it his 0 height
    result(i) = (sol(impact, 1) * 3.281) *.75;% .75 represents losses due to friction and
loss of energy during energy transfer

    %Capture the data of the best launch
    if bands == 10 && launch == 52.69
        ideal = sol;
    end
end

    %Plotting
    plot(Possible_Angles, result, '-o', 'Color', colors(k), 'DisplayName', [num2str(bands) '
Bands'])
end

%%Plot formating
grid on;
title('Catapult range Estimates')
xlabel('Launch Angle (Degrees)')
ylabel('Perdicted Range(Feet)')
legend('Location', 'best')

```

```
%Graph of the ideal launch
figure(2)
```

```
%Pull x and y from the solution
x = ideal(:, 1) * 3.281 * 0.75; %feet conversion and 0.75 loss
y = ideal(:, 3) * 3.281;      %feet conversion
```

```
%Plot
plot(x(y >= 0), y(y >= 0), 'LineWidth', 2, 'Color', 'b'); %Plot where its at or above the
ground
```

```
%Format of plot
grid on;
title('Ideal Launch');
xlabel('Perdicted Range(Feet)');
ylabel('Height(Feet)');
```

```
%function for differential equation for drag
function dydt = projectile_drag(~, y, m, Area)
    rho = 1.225; %Air density
    Cd = 0.5; %Drag coeffecient assumption
    g = 9.81; %Gravity constant
    v = sqrt(y(2)^2 + y(4)^2); %Velocity
    drag = 0.5 * rho * v^2 * Cd * Area; %Drag equation
    dydt = [y(2); -(drag/m)*(y(2)/v); y(4); -g-(drag/m)*(y(4)/v)]; %Projectile motion with
drag
end
```

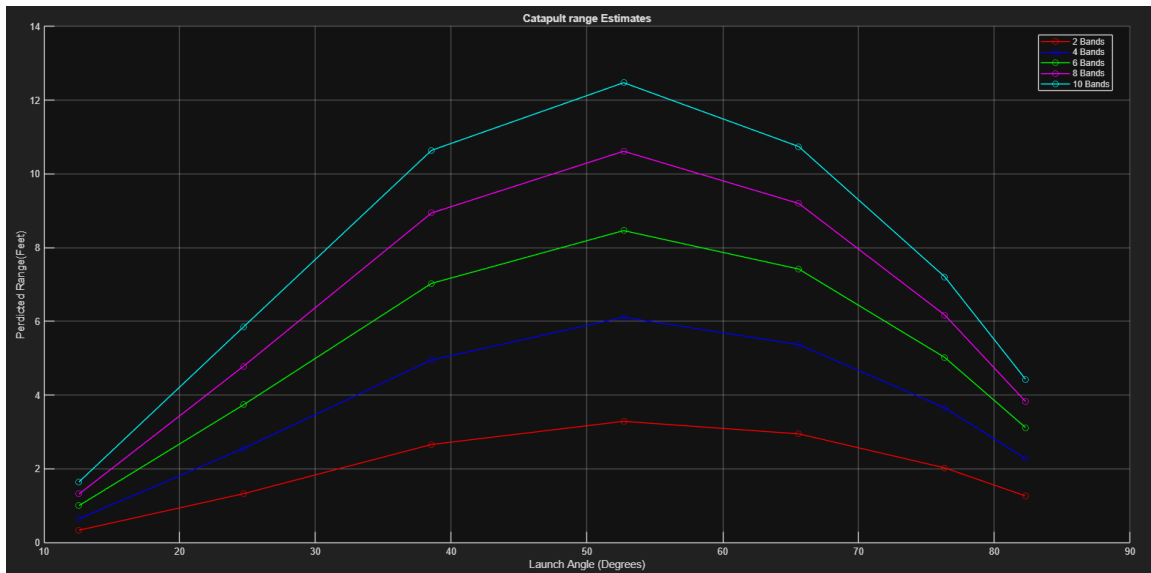


Figure 12: MATLAB graph showing the range various launch angles and numbers of rubberbands have

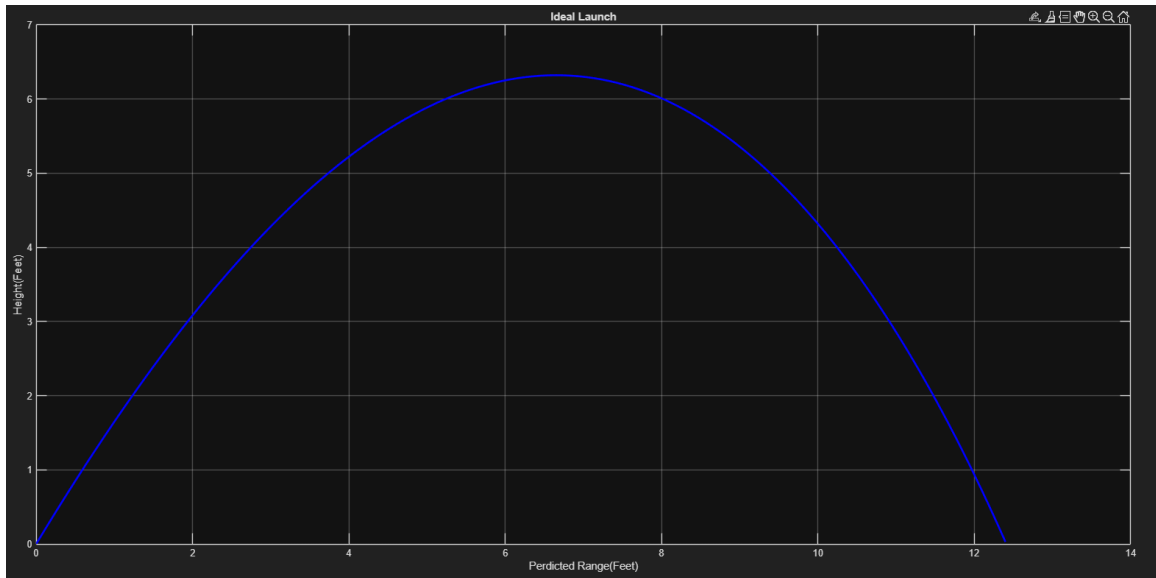


Figure 13: Trajectory of the Best Calculated Launch

Distance(meters)	Height(meters)	Distance(Feet) + Loss	Height(Feet) + Loss	Calculated initial velocity
0	0	0	0	8.62
0.052248167	0.068070196	0.085713118	0.111669157	
0.104496334	0.135159392	0.171426236	0.221728983	
0.156744501	0.201267588	0.257139354	0.330179478	
0.208992668	0.266394784	0.342852472	0.437020644	
0.261240835	0.33054098	0.42856559	0.542252478	
0.313489002	0.393706176	0.514278708	0.645874982	
0.365737169	0.455890372	0.599991826	0.747888156	
0.417985336	0.517093569	0.685704944	0.848291999	
0.470233503	0.577315765	0.771418062	0.947086512	
0.52248167	0.636556961	0.85713118	1.044271694	
0.574729837	0.694817157	0.942844298	1.139847546	
0.626978004	0.752096353	1.028557416	1.233814067	
0.679226171	0.808394549	1.114270533	1.326171257	
0.731474338	0.863711745	1.199983651	1.416919118	
0.783722505	0.918047941	1.285696769	1.506057647	
0.835970672	0.971403137	1.371409887	1.593586846	
0.888218839	1.023777333	1.457123005	1.679506715	
0.940467006	1.075170529	1.542836123	1.763817253	
0.992715173	1.125582725	1.628549241	1.846518461	
1.04496334	1.175013921	1.714262359	1.927610338	
1.097211507	1.223464117	1.799975477	2.007092885	
1.149459674	1.270933314	1.885688595	2.084966101	
1.201707841	1.31742151	1.971401713	2.161229986	
1.253956008	1.362928706	2.057114831	2.235884542	
1.306204175	1.407454902	2.142827949	2.308929766	
1.358452342	1.451000098	2.228541067	2.38036566	
1.410700509	1.493564294	2.314254185	2.450192224	
1.462948676	1.53514749	2.399967303	2.518409457	
1.515196843	1.575749686	2.485680421	2.58501736	
1.56744501	1.615370882	2.571393539	2.650015932	
1.619693177	1.654011078	2.657106657	2.713405174	
1.671941344	1.691670274	2.742819775	2.775185085	
1.724189511	1.72834847	2.828532893	2.835355665	
1.776437678	1.764045666	2.914246011	2.893916916	
1.828685845	1.798761862	3.000059129	2.950958925	

Figure 14: Snip of an Excel sheet giving the calculated trajectory of the catapult

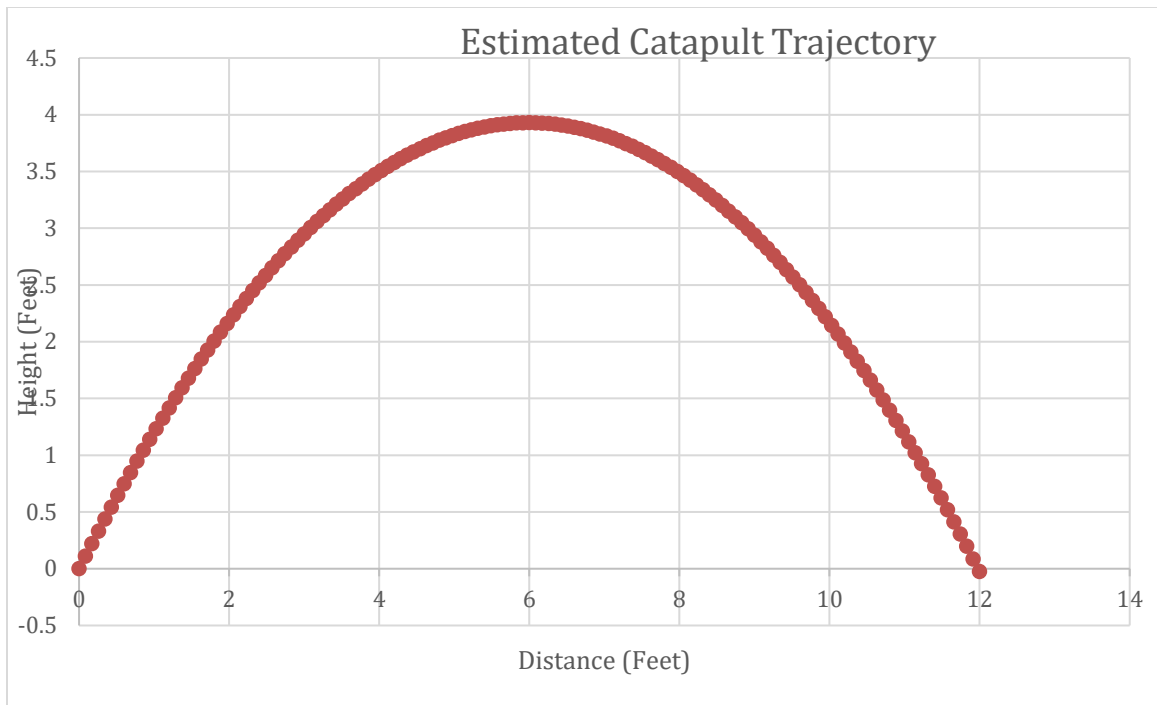


Figure 15: Excel graph of the catapults calculated trajectory

Appendix G: Solidworks Model

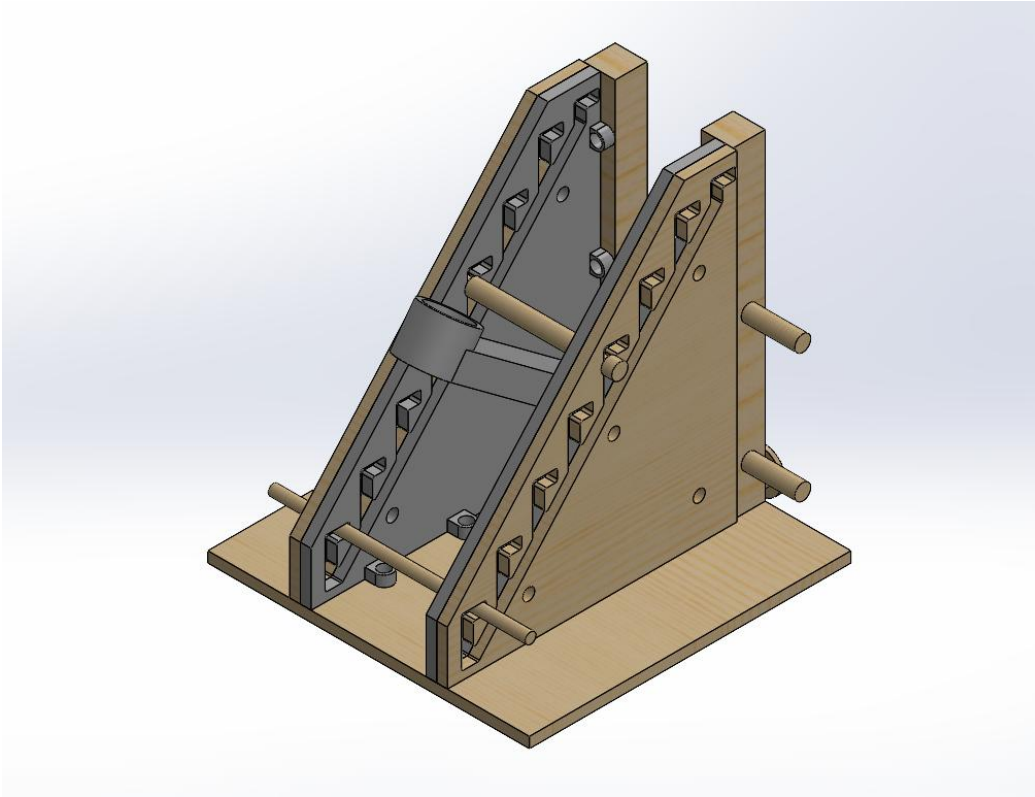
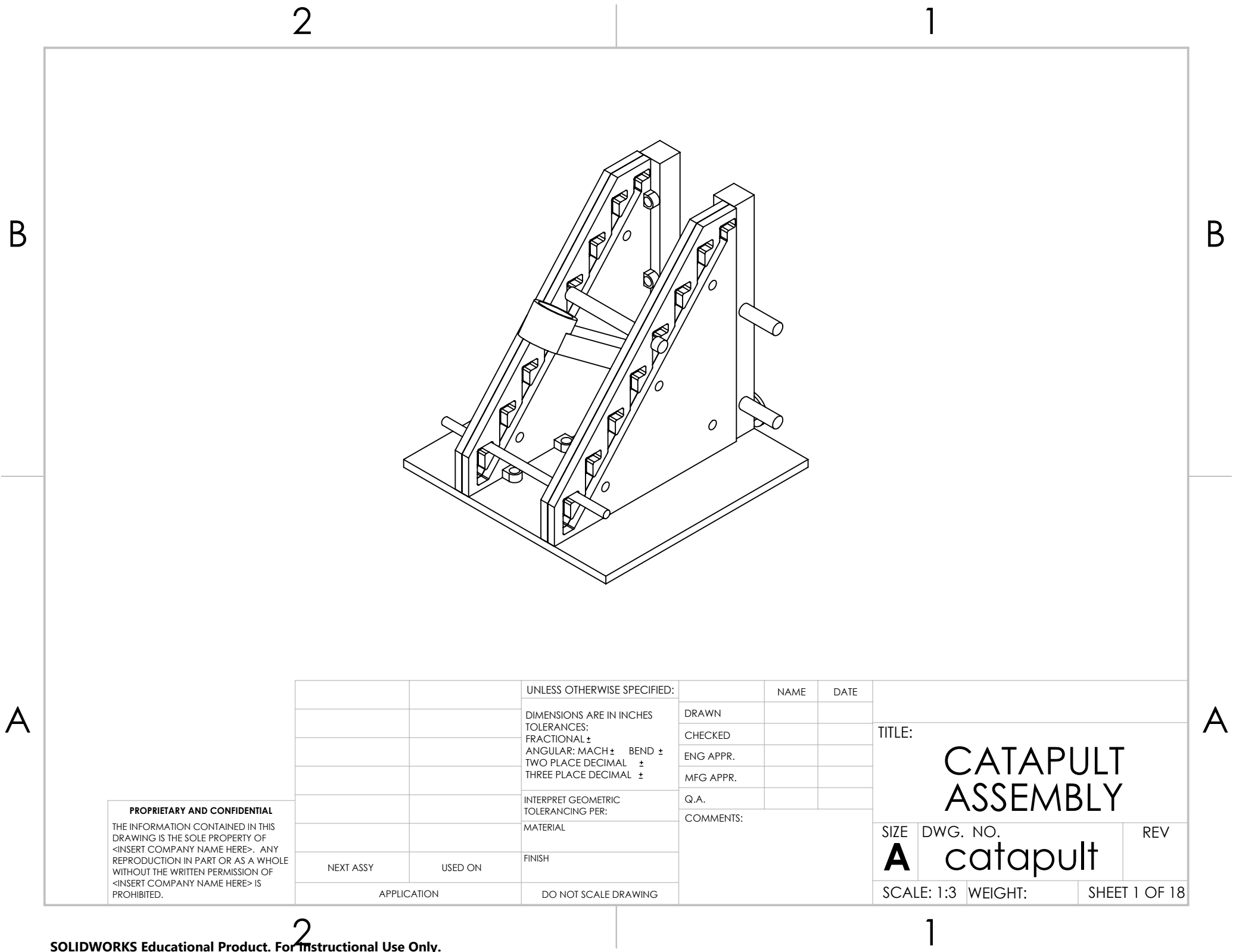


Figure 16: Final Catapult CAD Model

Some of the modeling tools used to make this part are extrusions and cuts. Each part of the catapult is relatively simple and flat. We also used fillets to remove any sharp corners that could become fracture points. This also makes use of coincident, concentric, tangent, and width mates to put all of the pieces into the complete model.

Appendix H: Drawing Package



A

A

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

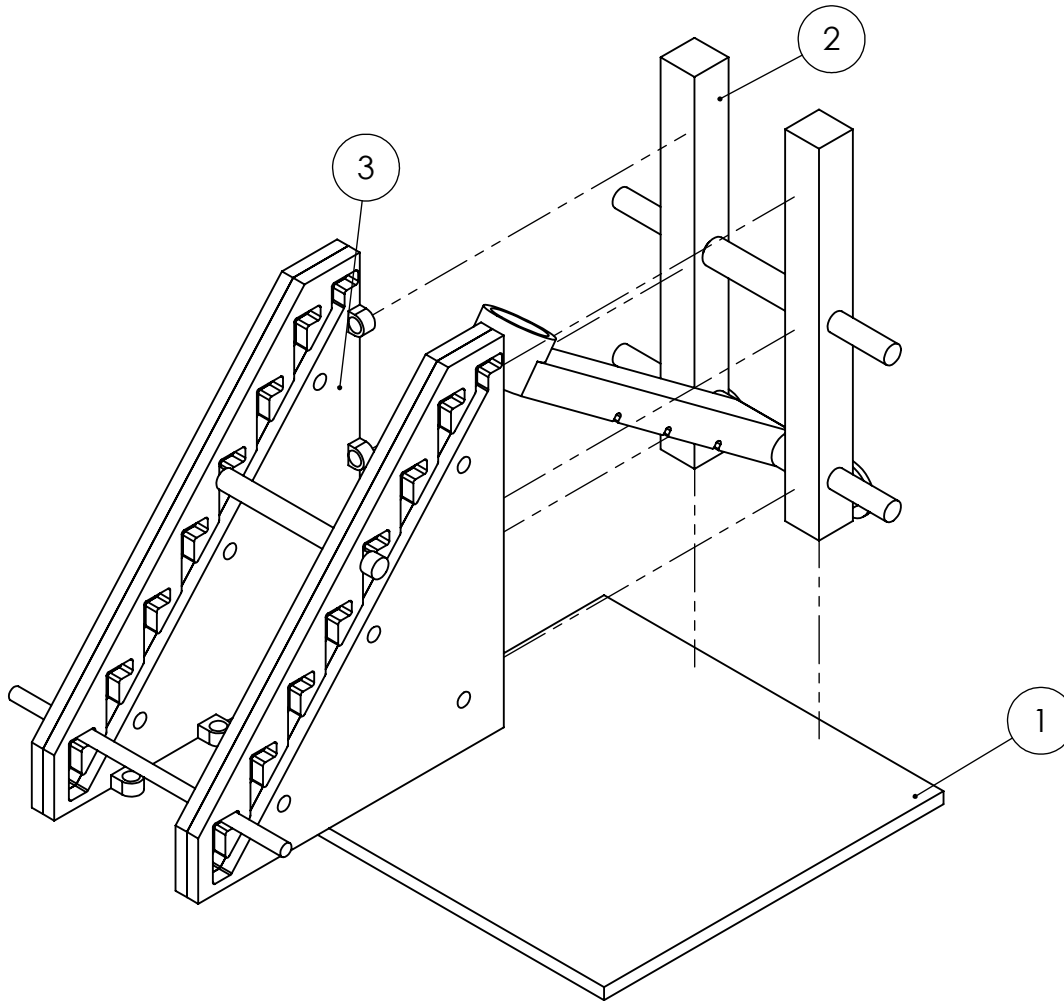
		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS ARE IN INCHES	DRAWN		
		TOLERANCES:	CHECKED		
		FRACTIONAL $\pm$	ENG APPR.		
		ANGULAR: MACH $\pm$ BEND $\pm$	MFG APPR.		
		TWO PLACE DECIMAL $\pm$	Q.A.		
		THREE PLACE DECIMAL $\pm$	COMMENTS:		
		INTERPRET GEOMETRIC TOLERANCING PER:			
		MATERIAL			
NEXT ASSY	USED ON	FINISH			
APPLICATION		DO NOT SCALE DRAWING			

TITLE:

CATAPULT
ASSEMBLY

SIZE	DWG. NO.	REV
A	catapult	

SCALE: 1:3	WEIGHT:	SHEET 1 OF 18
------------	---------	---------------



ITEMS TO BE CONNETED USING
SCREWS UNLESS OTHERWISE STATED
ITEM 2 TO BE GLUED AND
SCREWED TO ITEM 1

ITEM NO.	PART NUMBER	QTY.
1	BASE	1
2	ArmAssembly	1
3	StructureAssembly	1

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

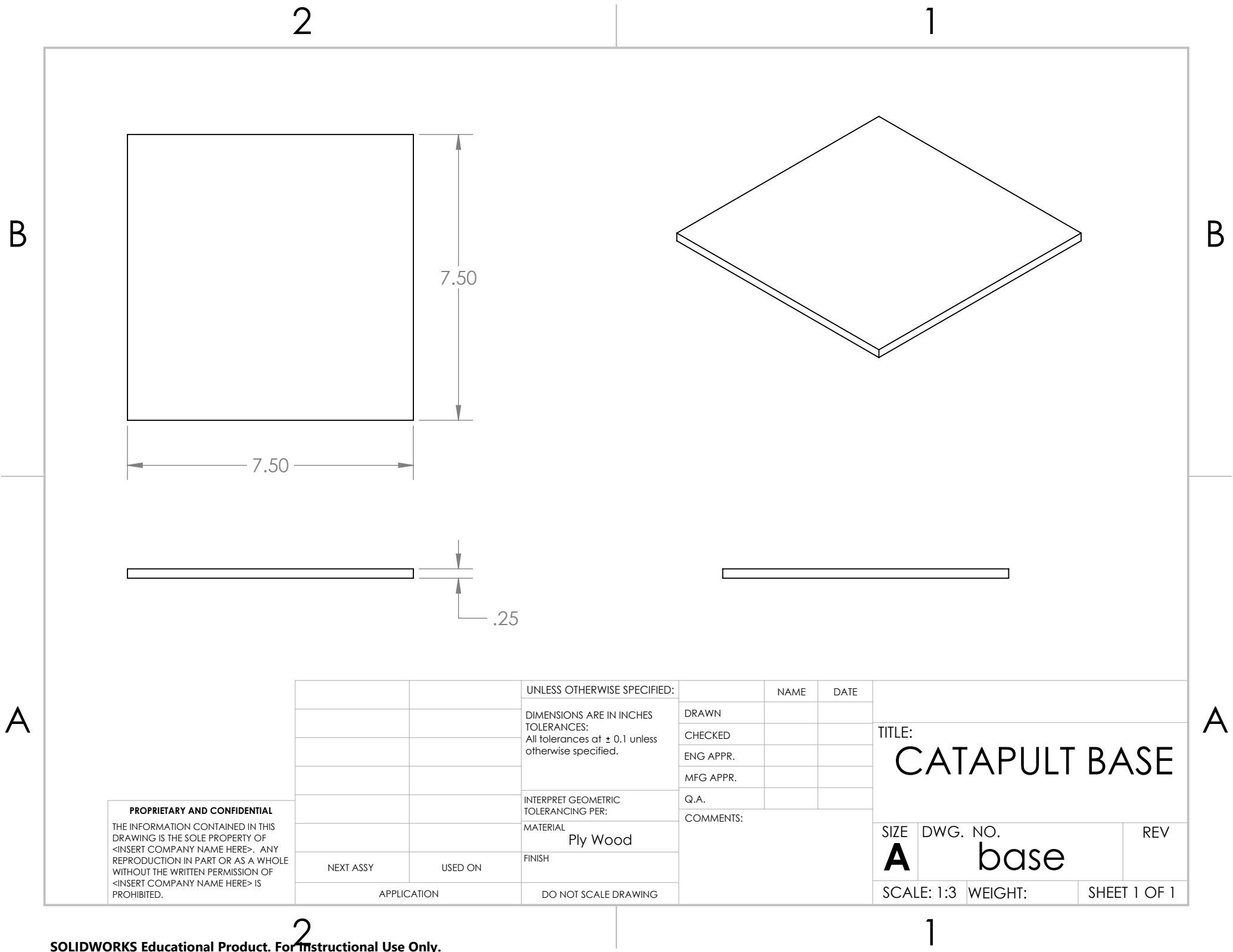
		UNLESS OTHERWISE SPECIFIED:
		DIMENSIONS ARE IN INCHES
		TOLERANCES:
		FRACTIONAL $\pm$
		ANGULAR: MACH $\pm$ BEND $\pm$
		TWO PLACE DECIMAL $\pm$
		THREE PLACE DECIMAL $\pm$
		INTERPRET GEOMETRIC TOLERANCING PER:
		MATERIAL
NEXT ASSY	USED ON	FINISH
APPLICATION		DO NOT SCALE DRAWING

	NAME	DATE
DRAWN		
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

TITLE:

CATAPULT
ASSEMBLY

SIZE	DWG. NO.	REV
A	catapult	
SCALE: 1:3	WEIGHT:	SHEET 2 OF 2



A

A

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:
		DIMENSIONS ARE IN INCHES TOLERANCES: All tolerances at ± 0.1 unless otherwise specified.
		INTERPRET GEOMETRIC TOLERANCING PER:
		MATERIAL Ply Wood
NEXT ASSY	USED ON	FINISH
APPLICATION		DO NOT SCALE DRAWING

	NAME	DATE
DRAWN		
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

TITLE:
CATAPULT BASE

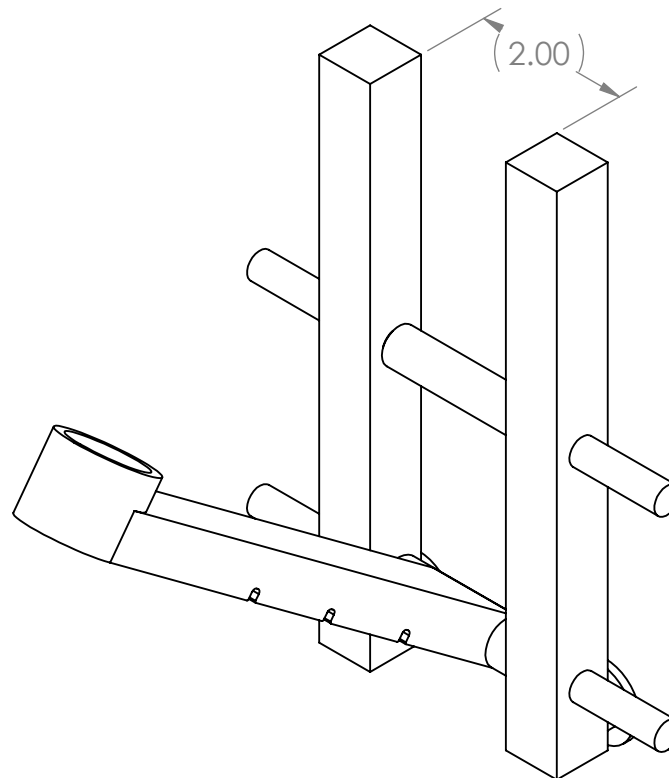
SIZE	DWG. NO.	REV
A	base	
SCALE: 1:3	WEIGHT:	SHEET 1 OF 1

2

1

B

B



A

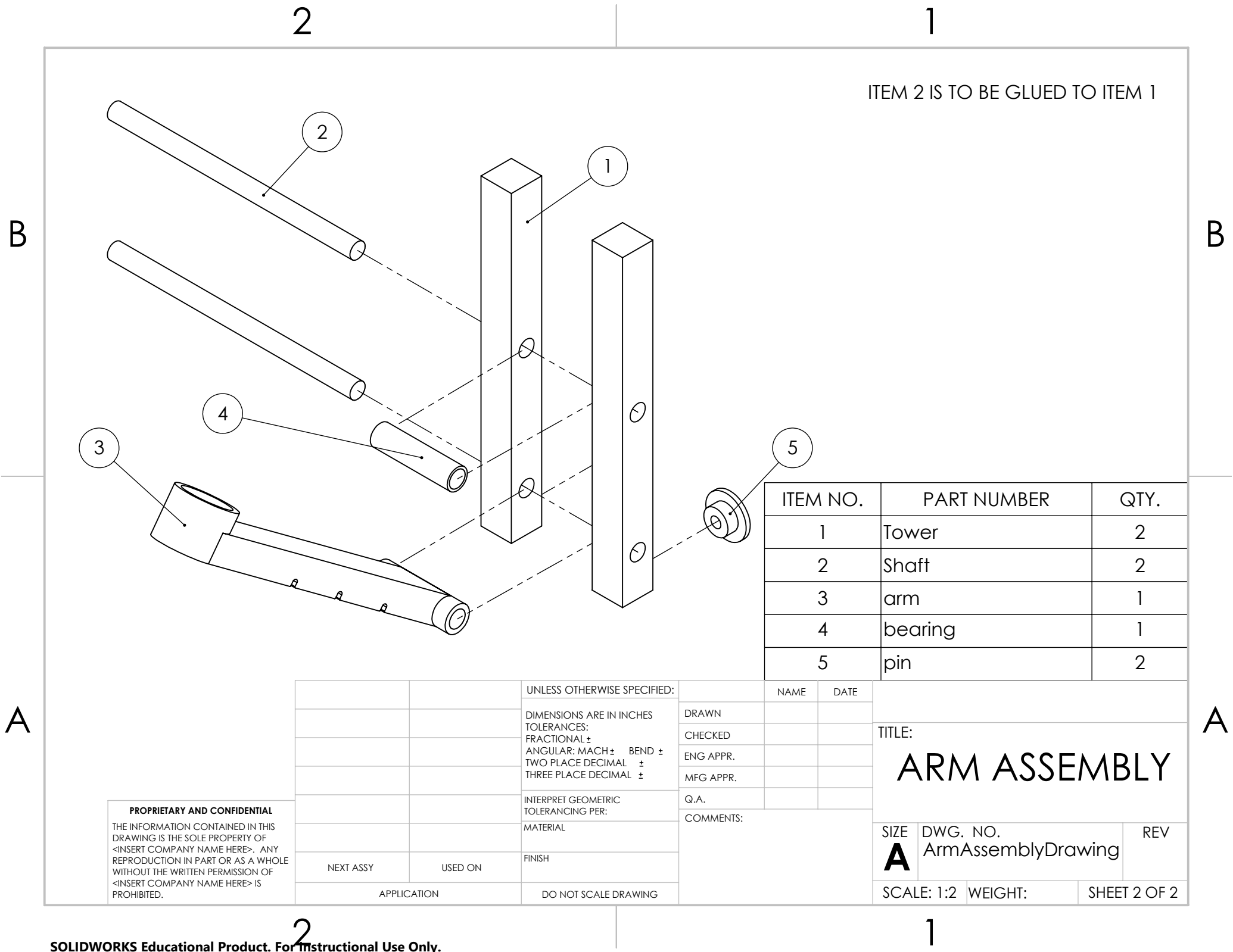
A

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TITLE: ARM ASSEMBLY			
		DIMENSIONS ARE IN INCHES	DRAWN						
		TOLERANCES:	CHECKED						
		FRACTIONAL ±	ENG APPR.						
		ANGULAR: MACH ± BEND ±	MFG APPR.						
		TWO PLACE DECIMAL ±	Q.A.			SIZE	DWG. NO.	REV	
		THREE PLACE DECIMAL ±	COMMENTS:			A	ArmAssemblyDrawing		
		INTERPRET GEOMETRIC TOLERANCING PER:				SCALE: 1:2	WEIGHT:	SHEET 1 OF 2	
		MATERIAL							
NEXT ASSY	USED ON	FINISH							
APPLICATION		DO NOT SCALE DRAWING							

2

1



ITEM 2 IS TO BE GLUED TO ITEM 1

ITEM NO.	PART NUMBER	QTY.
1	Tower	2
2	Shaft	2
3	arm	1
4	bearing	1
5	pin	2

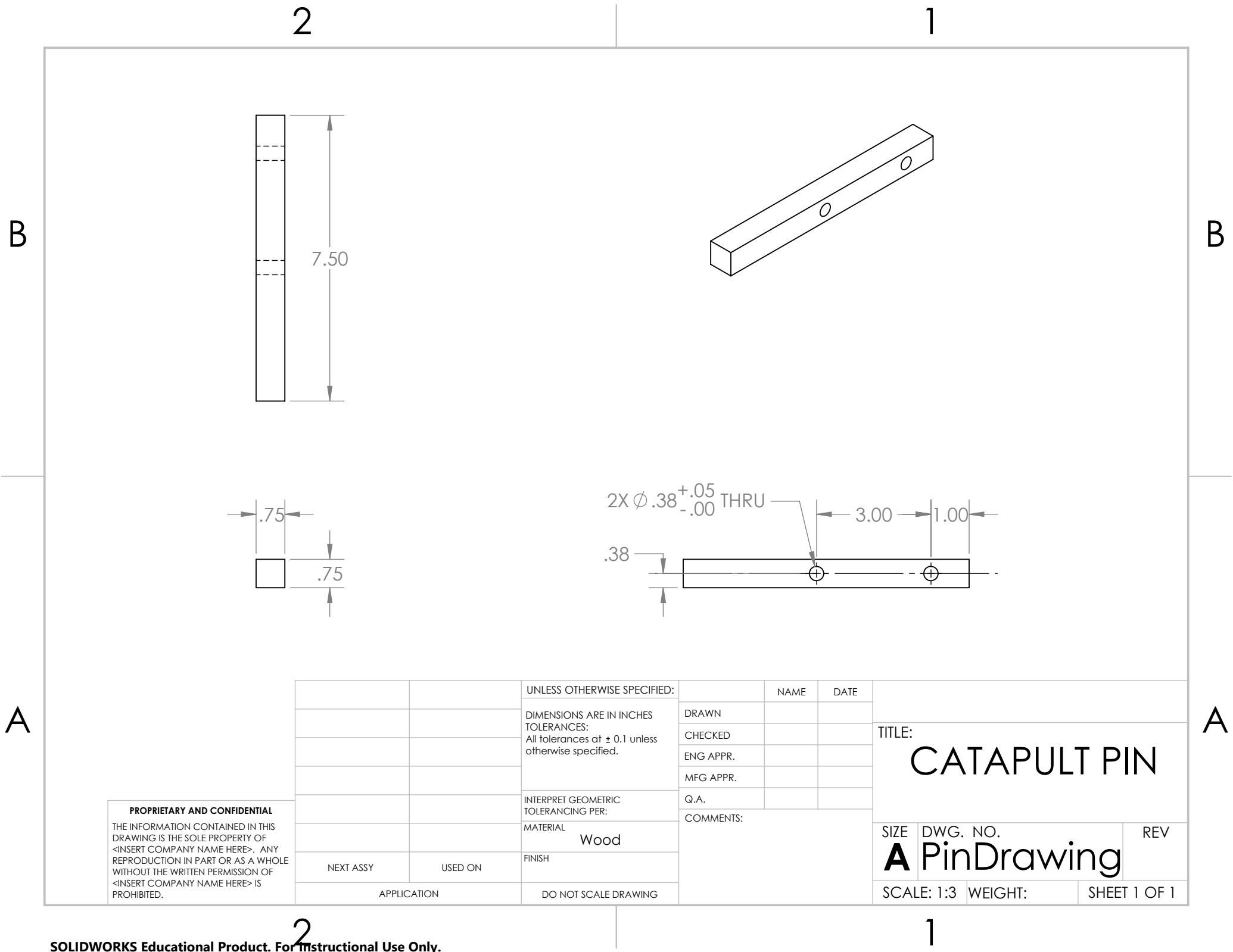
PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:
		DIMENSIONS ARE IN INCHES
		TOLERANCES:
		FRACTIONAL ±
		ANGULAR: MACH ± BEND ±
		TWO PLACE DECIMAL ±
		THREE PLACE DECIMAL ±
		INTERPRET GEOMETRIC TOLERANCING PER:
		MATERIAL
NEXT ASSY	USED ON	FINISH
APPLICATION		DO NOT SCALE DRAWING

NAME	DATE
DRAWN	
CHECKED	
ENG APPR.	
MFG APPR.	
Q.A.	
COMMENTS:	

TITLE:
ARM ASSEMBLY

SIZE	DWG. NO.	REV
A	ArmAssemblyDrawing	
SCALE: 1:2	WEIGHT:	SHEET 2 OF 2



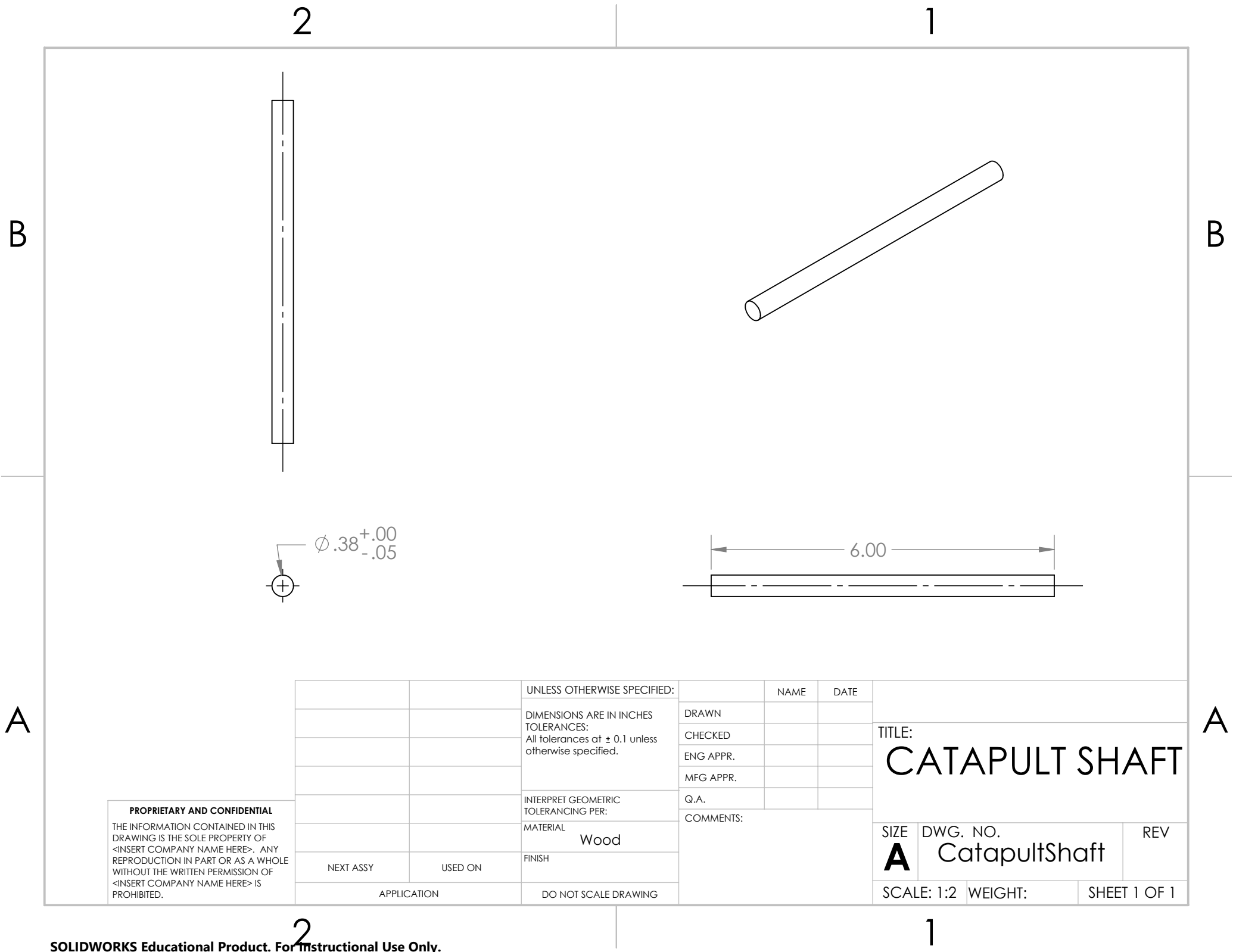
A

A

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS ARE IN INCHES TOLERANCES: All tolerances at ± 0.1 unless otherwise specified.	DRAWN		
			CHECKED		
			ENG APPR.		
			MFG APPR.		
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.		
		MATERIAL Wood	COMMENTS:		
NEXT ASSY	USED ON	FINISH			
APPLICATION		DO NOT SCALE DRAWING			

TITLE: CATAPULT PIN			SIZE	DWG. NO.	REV
			A PinDrawing		
SCALE: 1:3		WEIGHT:	SHEET 1 OF 1		

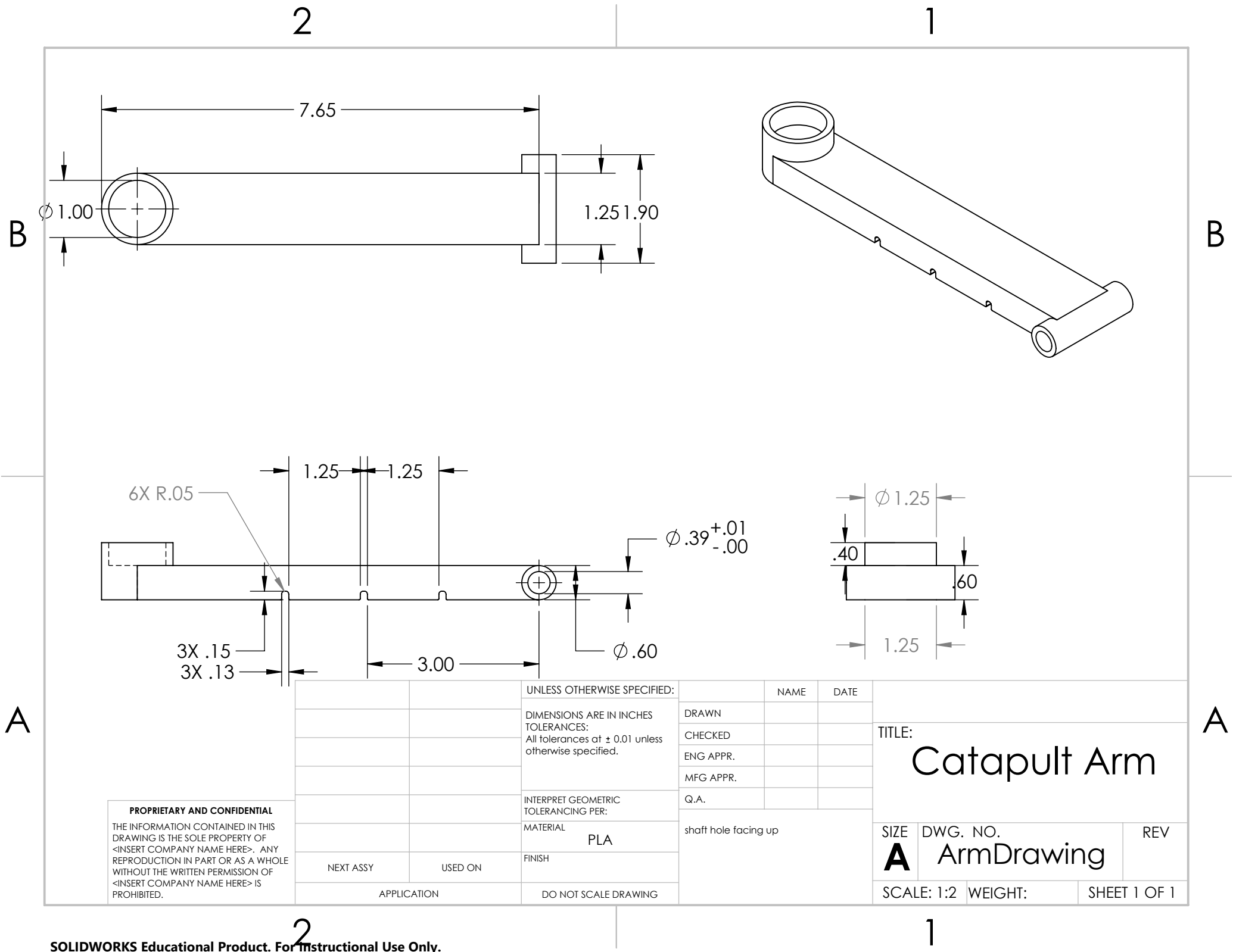


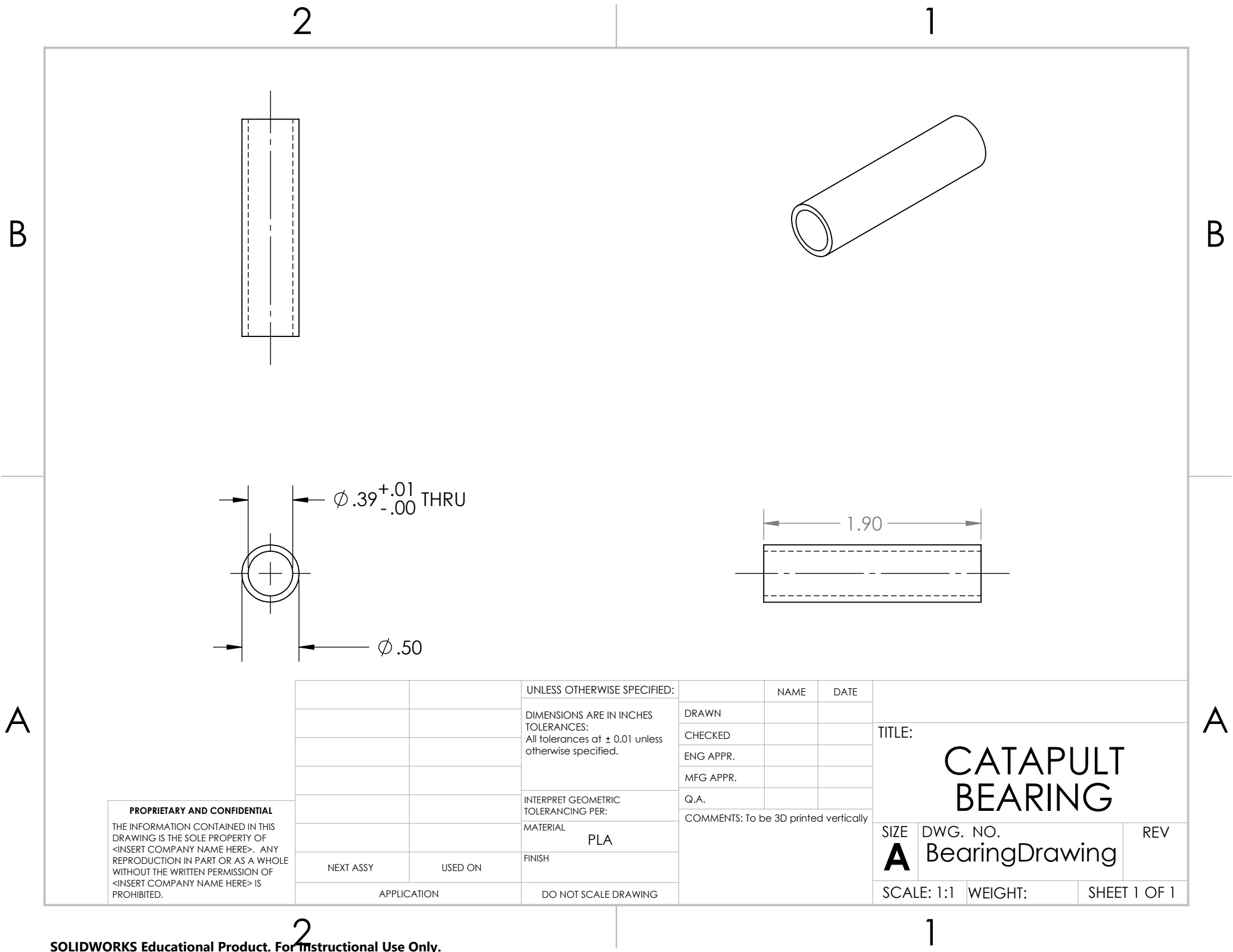
PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:
		DIMENSIONS ARE IN INCHES TOLERANCES: All tolerances at ± 0.1 unless otherwise specified.
		INTERPRET GEOMETRIC TOLERANCING PER:
		MATERIAL Wood
NEXT ASSY	USED ON	FINISH
APPLICATION		DO NOT SCALE DRAWING

	NAME	DATE
DRAWN		
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

TITLE: CATAPULT SHAFT		
SIZE A	DWG. NO. CatapultShaft	REV
SCALE: 1:2	WEIGHT:	SHEET 1 OF 1





B

B

A

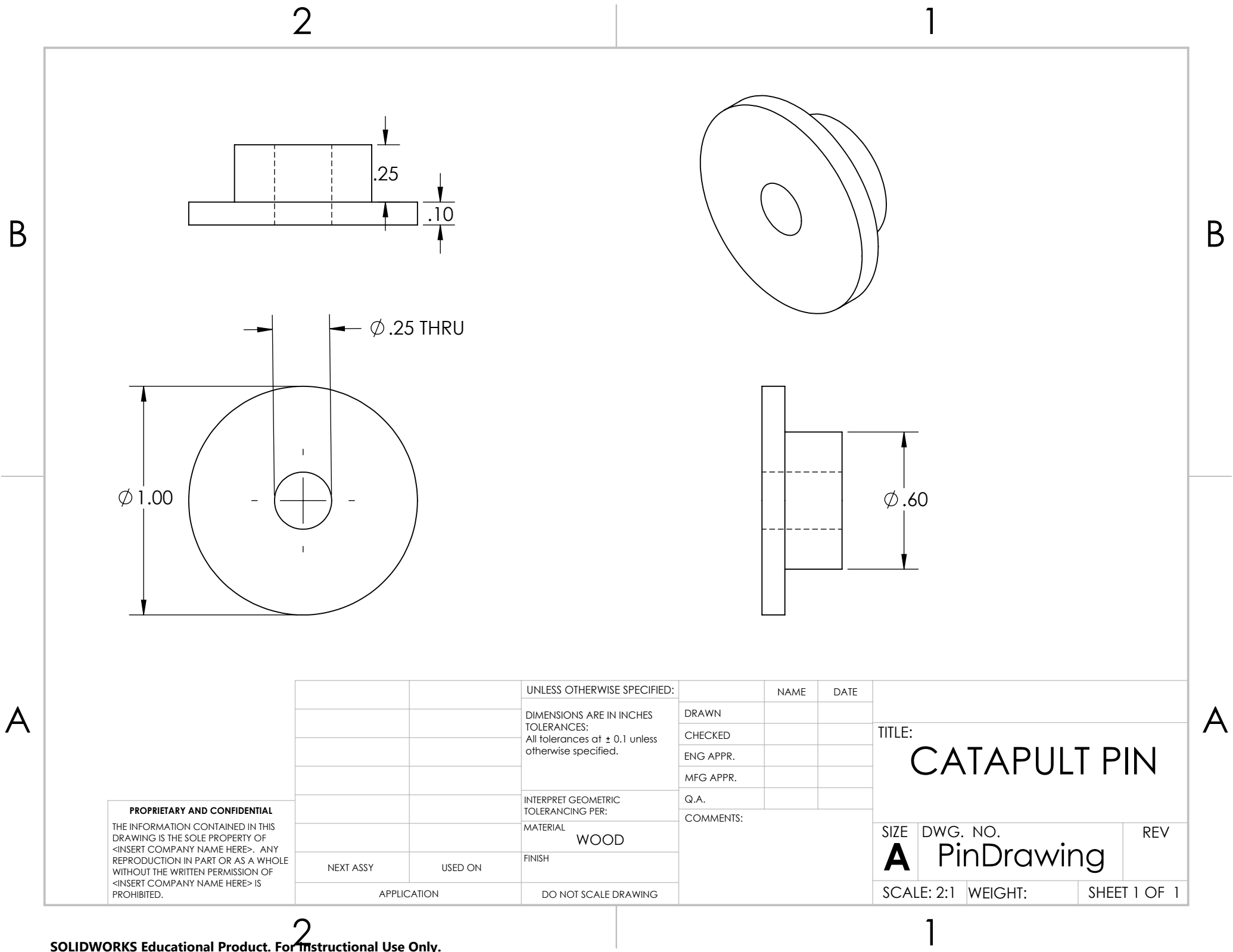
A

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:
		DIMENSIONS ARE IN INCHES TOLERANCES: All tolerances at ± 0.01 unless otherwise specified.
		INTERPRET GEOMETRIC TOLERANCING PER:
		MATERIAL PLA
NEXT ASSY	USED ON	FINISH
APPLICATION		DO NOT SCALE DRAWING

	NAME	DATE
DRAWN		
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS: To be 3D printed vertically		

TITLE: CATAPULT BEARING		
SIZE A	DWG. NO. BearingDrawing	REV
SCALE: 1:1	WEIGHT:	SHEET 1 OF 1



A

A

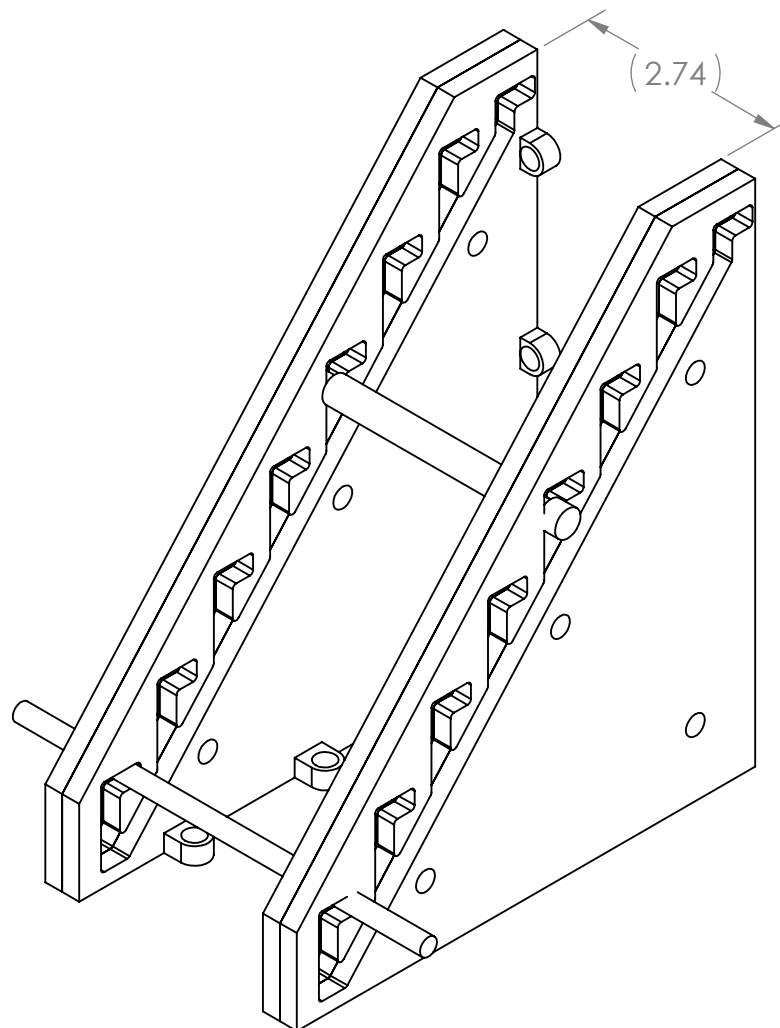
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.				UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TITLE: CATAPULT PIN	
				DIMENSIONS ARE IN INCHES TOLERANCES: All tolerances at ± 0.1 unless otherwise specified.	DRAWN				
					CHECKED				
					ENG APPR.				
					MFG APPR.				
				INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.			COMMENTS:	
		MATERIAL	WOOD						
		FINISH							
NEXT ASSY	USED ON	FINISH							
APPLICATION		DO NOT SCALE DRAWING		SCALE: 2:1 WEIGHT: SHEET 1 OF 1					
				SIZE A DWG. NO. PinDrawing REV					

2

1

B

B



A

A

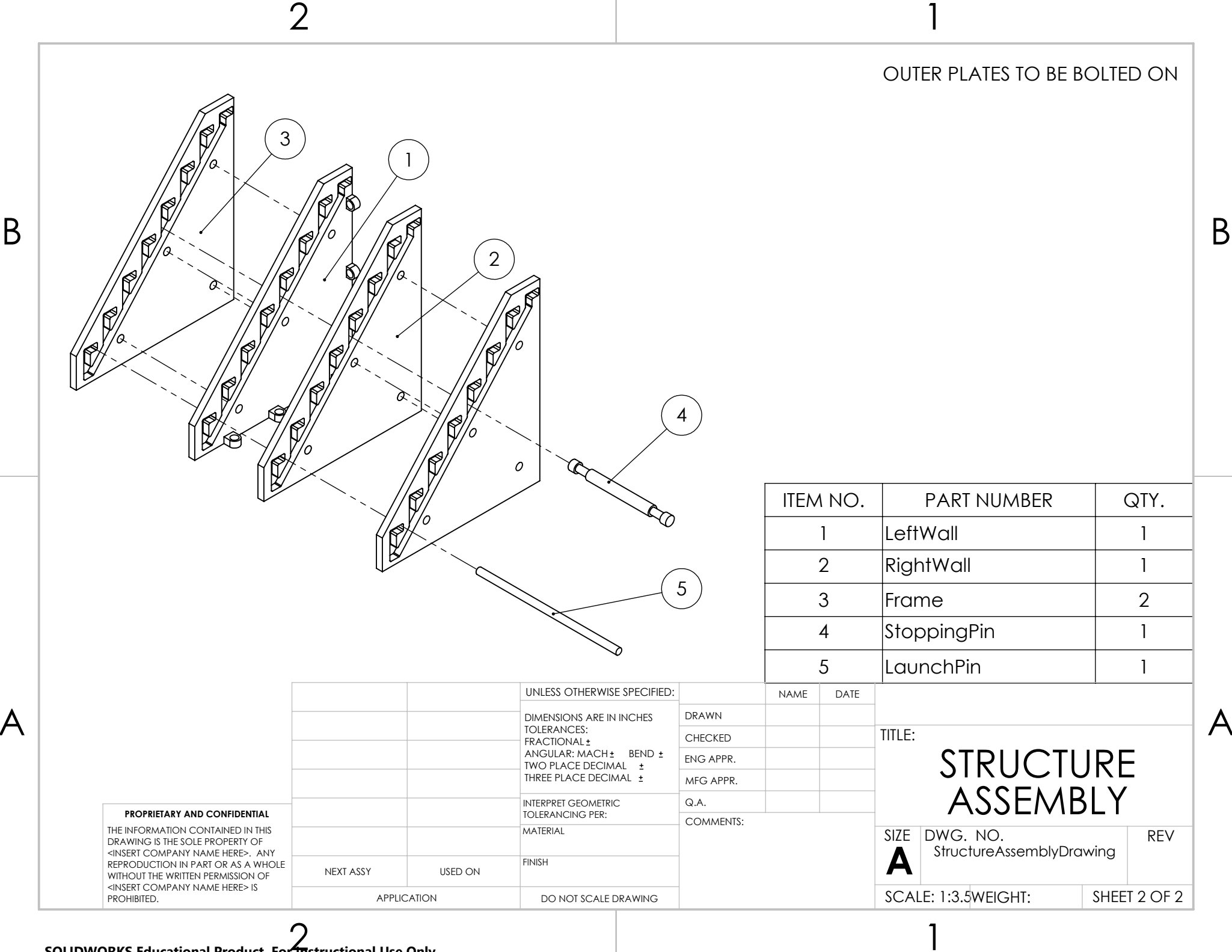
PROPRIETARY AND CONFIDENTIAL

THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TITLE: STRUCTURE ASSEMBLY		
		DIMENSIONS ARE IN INCHES	DRAWN					
		TOLERANCES:	CHECKED					
		FRACTIONAL ±	ENG APPR.					
		ANGULAR: MACH ± BEND ±	MFG APPR.					
		TWO PLACE DECIMAL ±	Q.A.			SIZE A		
		THREE PLACE DECIMAL ±	COMMENTS:					
		INTERPRET GEOMETRIC TOLERANCING PER:				DWG. NO.		REV
		MATERIAL				StructureAssemblyDrawing		
		FINISH				SCALE: 1:2		WEIGHT:
NEXT ASSY	USED ON					SHEET 1 OF 2		
APPLICATION		DO NOT SCALE DRAWING						

2

1



OUTER PLATES TO BE BOLTED ON

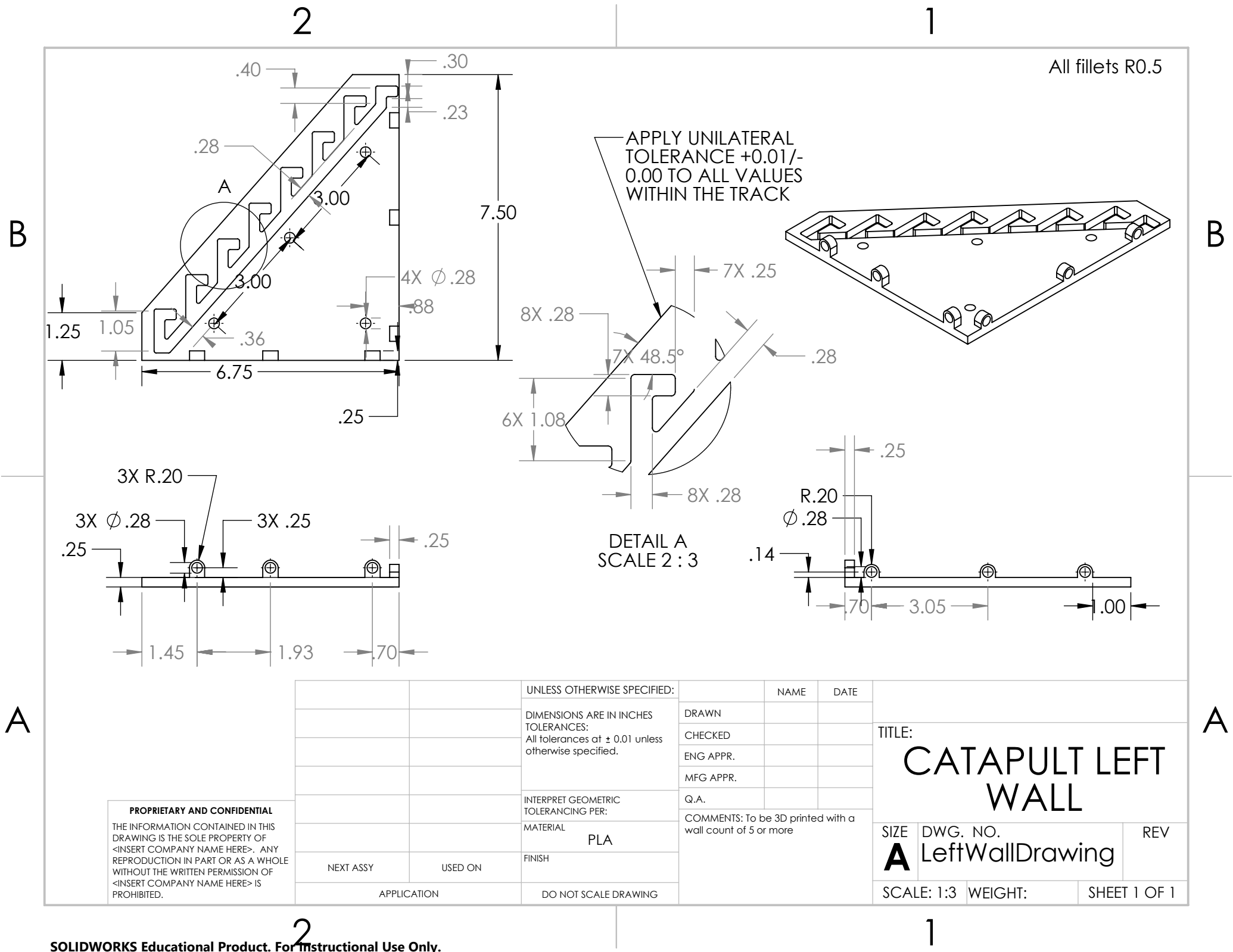
ITEM NO.	PART NUMBER	QTY.
1	LeftWall	1
2	RightWall	1
3	Frame	2
4	StoppingPin	1
5	LaunchPin	1

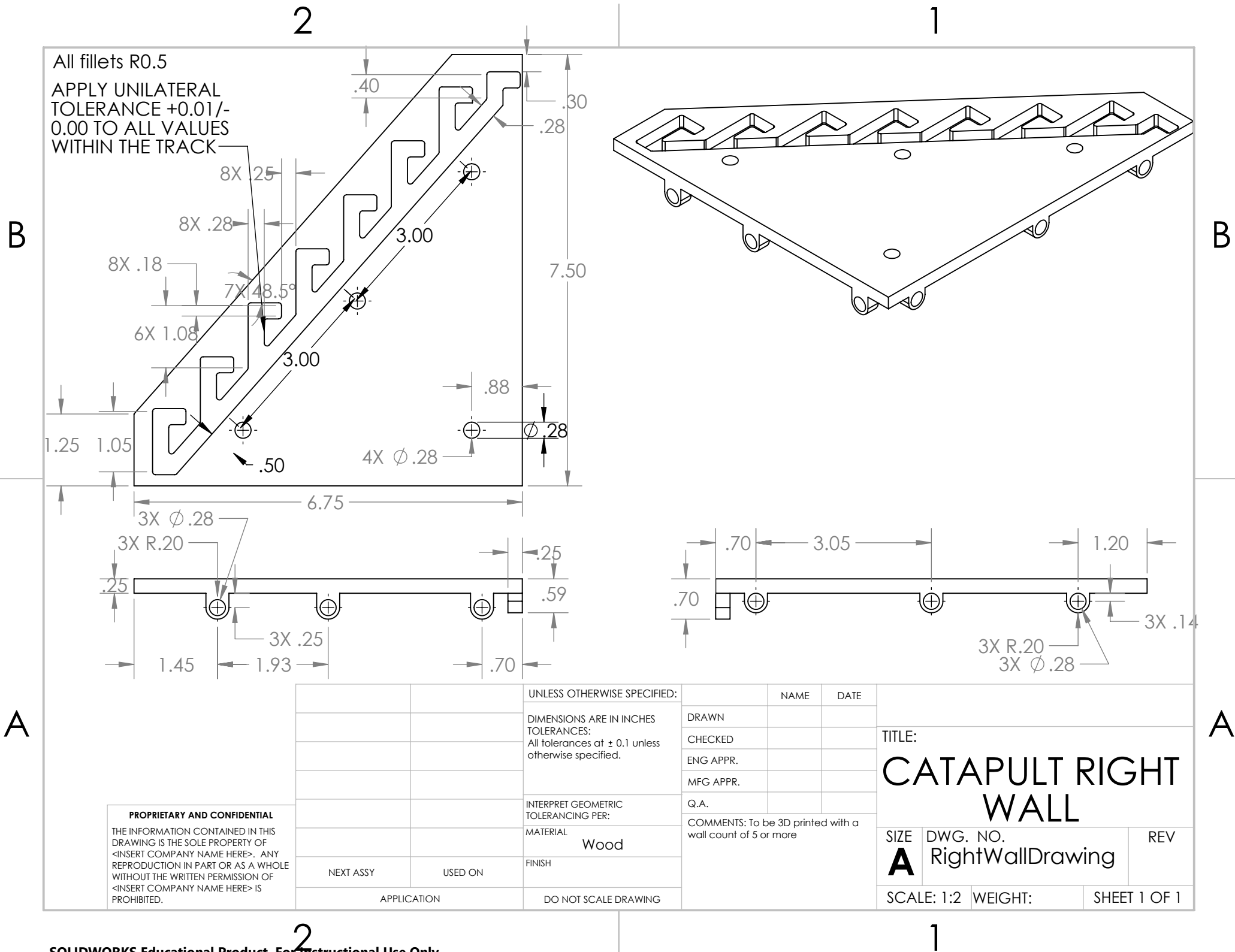
PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:
		DIMENSIONS ARE IN INCHES
		TOLERANCES:
		FRACTIONAL ±
		ANGULAR: MACH ± BEND ±
		TWO PLACE DECIMAL ±
		THREE PLACE DECIMAL ±
		INTERPRET GEOMETRIC
		TOLERANCING PER:
		MATERIAL
		FINISH
NEXT ASSY	USED ON	
APPLICATION		DO NOT SCALE DRAWING

NAME	DATE
DRAWN	
CHECKED	
ENG APPR.	
MFG APPR.	
Q.A.	
COMMENTS:	

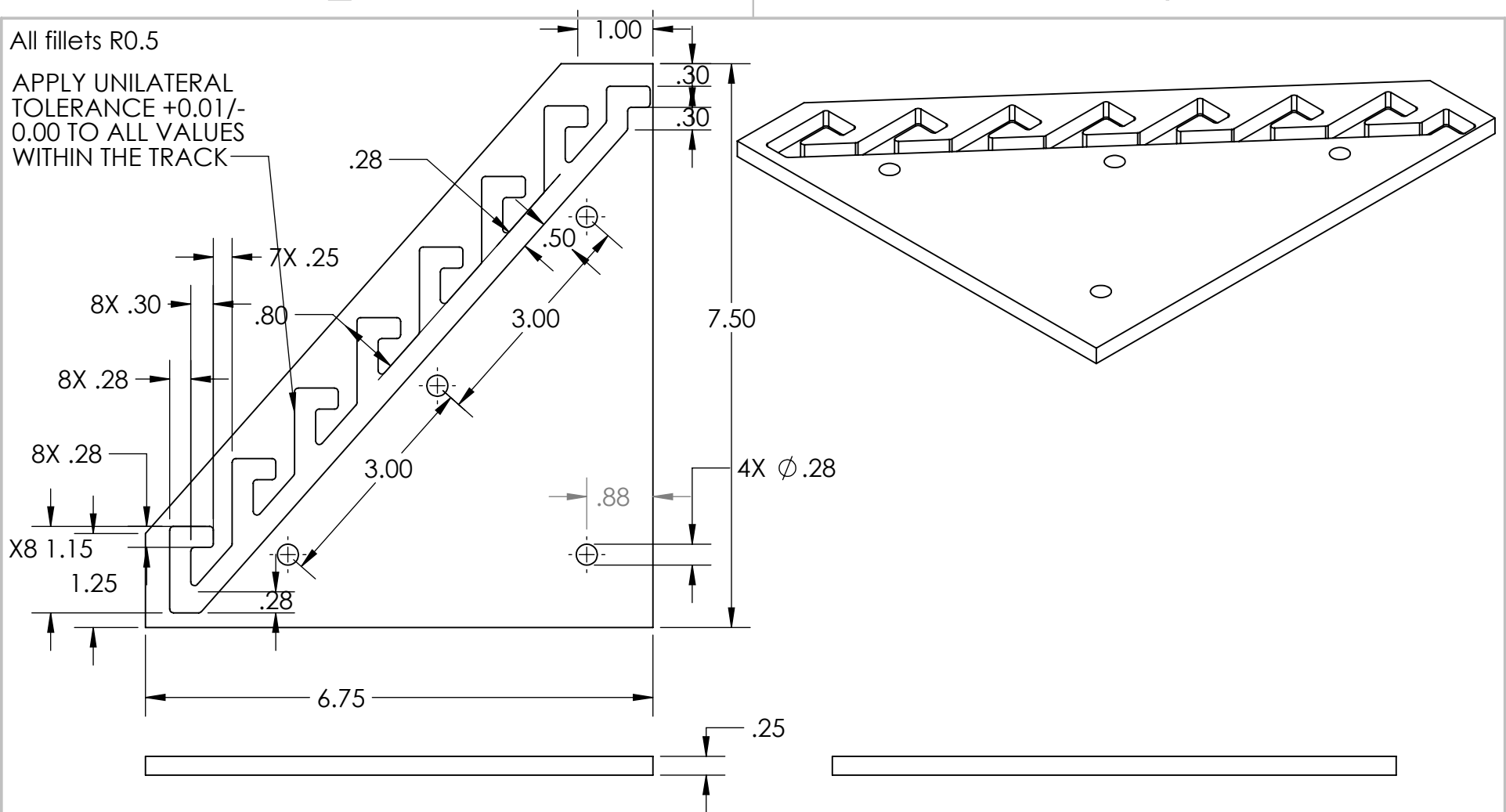
TITLE: STRUCTURE ASSEMBLY		
SIZE A	DWG. NO. StructureAssemblyDrawing	REV
SCALE: 1:3.5		WEIGHT: SHEET 2 OF 2





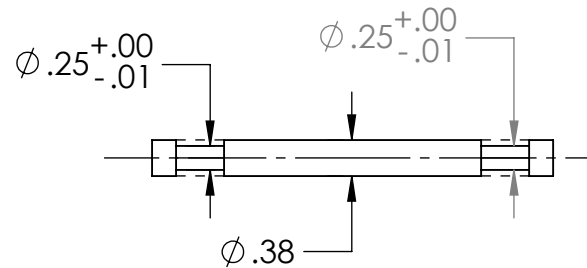
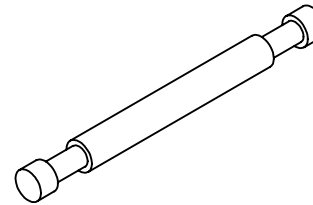
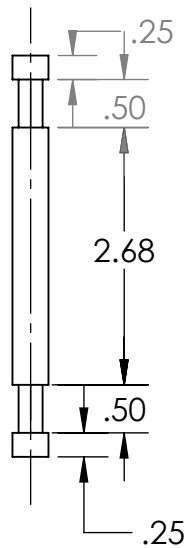
1

APPLY UNILATERAL
TOLERANCE +0.01/-
0.00 TO ALL VALUES
WITHIN THE TRACK—



THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

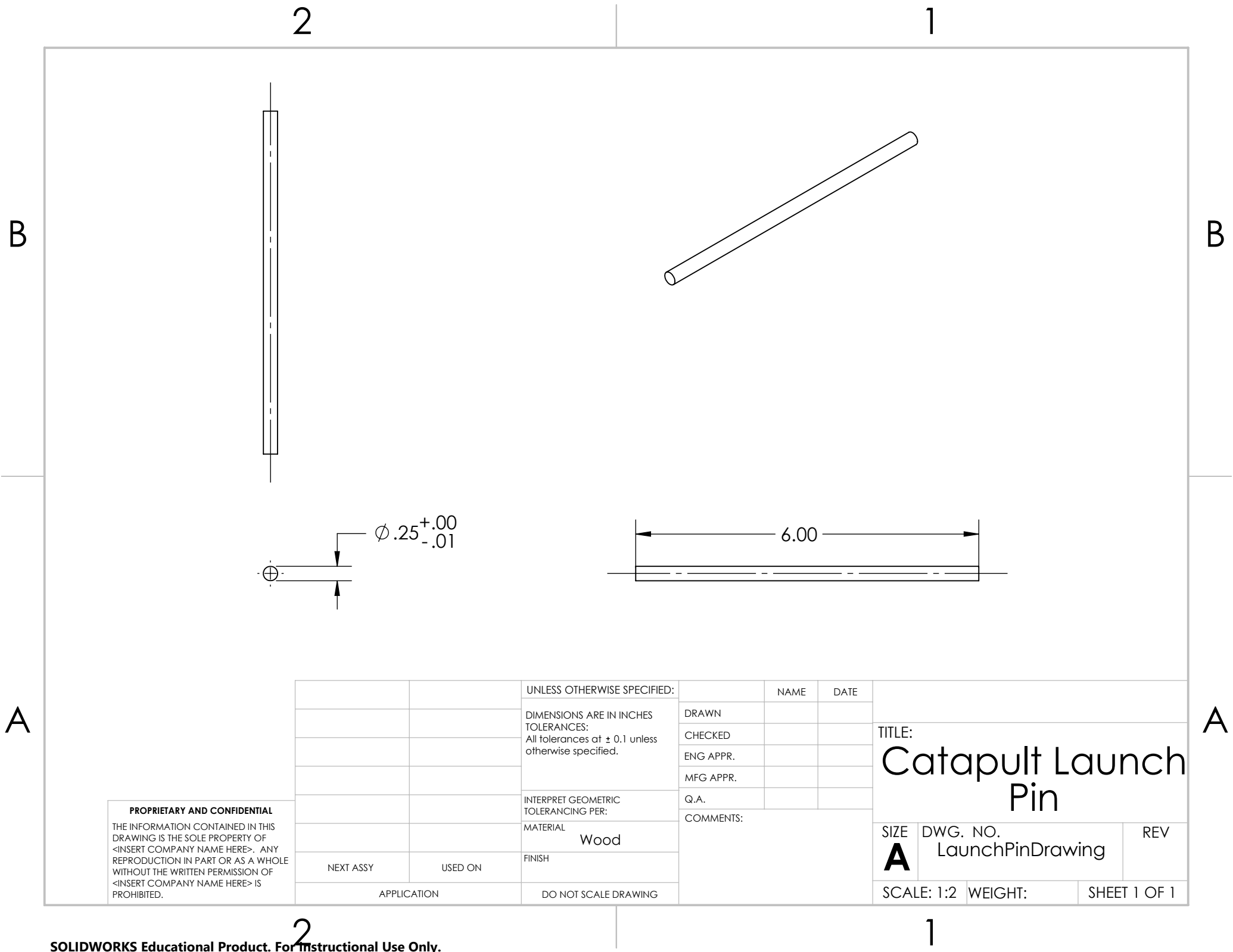
		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TITLE: CATAPULT FRAME				
		DIMENSIONS ARE IN INCHES TOLERANCES: All tolerances at ± 0.01 unless otherwise specified.	DRAWN							
			CHECKED							
			ENG APPR.							
			MFG APPR.							
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.			SIZE A DWG. NO. FrameDrawing REV				
		MATERIAL Wood	COMMENTS:							
NEXT ASSY	USED ON	FINISH								
APPLICATION		DO NOT SCALE DRAWING	SCALE: 1:2				WEIGHT:		SHEET 1 OF 1	



PROPRIETARY AND CONFIDENTIAL

THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TITLE: CATAPULT STOPPING PIN		
		DIMENSIONS ARE IN INCHES TOLERANCES: All tolerances at ± 0.1 unless otherwise specified.	DRAWN					
			CHECKED					
			ENG APPR.					
			MFG APPR.					
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.			SIZE A DWG. NO. StoppingPinDrawing REV		
		MATERIAL Wood	COMMENTS:					
NEXT ASSY	USED ON	FINISH						
APPLICATION		DO NOT SCALE DRAWING				SCALE: 1:2 WEIGHT: SHEET 1 OF 1		



A

A

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS ARE IN INCHES TOLERANCES: All tolerances at ± 0.1 unless otherwise specified.	DRAWN		
			CHECKED		
			ENG APPR.		
			MFG APPR.		
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.		
		MATERIAL Wood	COMMENTS:		
NEXT ASSY	USED ON	FINISH			
APPLICATION		DO NOT SCALE DRAWING			

TITLE: Catapult Launch Pin			
SIZE A	DWG. NO. LaunchPinDrawing		REV
SCALE: 1:2		WEIGHT:	SHEET 1 OF 1